Exploratory Data Analysis

STOR 320.(01 OR 02) Group PLACE_GROUP_NUMBER_HERE (Ex: STOR 320.01 Group 12)

March 23, 2026

```r
knitr::opts_chunk$set(echo = TRUE)
suppressWarnings(library(tidyverse))
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
suppressWarnings(library(janitor))
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
suppressWarnings(library(stringr))
suppressWarnings(library(DataExplorer))
suppressWarnings(library(skimr))
suppressWarnings(library(corrr))
```

```
##
## Attaching package: 'corrr'
##
## The following object is masked from 'package:skimr':
##
##     focus
```

```r
suppressWarnings(library(tidyverse))
suppressWarnings(library(car))
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```r
suppressWarnings(library(gt))
library(funModeling)
```

```
## Warning: package 'funModeling' was built under R version 4.5.2

## Loading required package: Hmisc

## Warning: package 'Hmisc' was built under R version 4.5.2

##
## Attaching package: 'Hmisc'
##
## The following objects are masked from 'package:gt':
##
##     html, latex
##
## The following objects are masked from 'package:dplyr':
##
##     src, summarize
##
## The following objects are masked from 'package:base':
##
##     format.pval, units
##
## funModeling v.1.9.6 :)
## Examples and tutorials at livebook.datascienceheroes.com
##  / Now in Spanish: librovivodecienciadedatos.ai
```

```r
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.5.2
```

```r
# Import Data Below
loan_dataset_full <- read_csv("../Raw Data/lc_loan.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 887379 Columns: 74
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (22): term, grade, sub_grade, emp_title, emp_length, home_ownership, ver...
## dbl (49): id, member_id, loan_amnt, funded_amnt, funded_amnt_inv, int_rate, ...
## lgl  (3): annual_inc_joint, dti_joint, verification_status_joint
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Subset of actual dataset of interest:

```r
actual_data <- loan_dataset_full |>
  select(annual_inc, int_rate, loan_amnt, emp_length, dti,
         addr_state, grade, purpose, home_ownership, term) |>
  janitor::clean_names() |>
  mutate(across(where(is.character), as.factor))

glimpse(actual_data)
```

```
## Rows: 887,379
## Columns: 10
## $ annual_inc     <dbl> 24000.00, 30000.00, 12252.00, 49200.00, 80000.00, 36000~
## $ int_rate       <dbl> 10.65, 15.27, 15.96, 13.49, 12.69, 7.90, 15.96, 18.64, ~
## $ loan_amnt      <dbl> 5000, 2500, 2400, 10000, 3000, 5000, 7000, 3000, 5600, ~
## $ emp_length     <fct> 10+ years, < 1 year, 10+ years, 10+ years, 1 year, 3 ye~
## $ dti            <dbl> 27.65, 1.00, 8.72, 20.00, 17.94, 11.20, 23.51, 5.35, 5.~
## $ addr_state     <fct> AZ, GA, IL, CA, OR, AZ, NC, CA, CA, TX, AZ, CA, VA, IL,~
## $ grade          <fct> B, C, C, C, B, A, C, E, F, B, C, B, C, B, B, D, C, A, B~
## $ purpose        <fct> credit_card, car, small_business, other, other, wedding~
## $ home_ownership <fct> RENT, RENT, RENT, RENT, RENT, RENT, RENT, RENT, OWN, RE~
## $ term           <fct> 36 months, 60 months, 36 months, 36 months, 60 months, ~
```

```r
summary(actual_data)
```

```
##    annual_inc         int_rate       loan_amnt         emp_length
##  Min.   :      0   Min.   : 5.32   Min.   :  500   10+ years:291569
##  1st Qu.:  45000   1st Qu.: 9.99   1st Qu.: 8000   2 years  : 78870
##  Median :  65000   Median :12.99   Median :13000   < 1 year : 70605
##  Mean   :  75028   Mean   :13.25   Mean   :14755   3 years  : 70026
##  3rd Qu.:  90000   3rd Qu.:16.20   3rd Qu.:20000   1 year   : 57095
##  Max.   :9500000   Max.   :28.99   Max.   :35000   5 years  : 55704
##  NA's   :4                                         (Other)  :263510
##       dti            addr_state     grade                     purpose
##  Min.   :   0.00   CA     :129517   A:148202   debt_consolidation:524215
##  1st Qu.:  11.91   NY     : 74086   B:254535   credit_card       :206182
##  Median :  17.65   TX     : 71138   C:245860   home_improvement  : 51829
##  Mean   :  18.16   FL     : 60935   D:139542   other             : 42894
##  3rd Qu.:  23.95   IL     : 35476   E: 70705   major_purchase    : 17277
##  Max.   :9999.00   NJ     : 33256   F: 23046   small_business    : 10377
##                    (Other):482971   G:  5489   (Other)           : 34605
##   home_ownership          term
```

```
##  ANY     :     3   36 months:621125
##  MORTGAGE:443557   60 months:266254
##  NONE    :    50
##  OTHER   :   182
##  OWN     : 87470
##  RENT    :356117
##
```

###Statement of the business problem

**Redo

Despite their best efforts, LendingClub is approving too many borrowers who default, scaring away potential investors. LendingClub has hired us to analyze their loan data and identify patterns that could help them better predict which borrowers are likely to default. By understanding the relationships between borrower characteristics, loan attributes, and loan performance, we can provide actionable insights to improve LendingClub's credit risk assessment and reduce default rate

Data Dictionary

RESEARCH MORE*

Variable	Type	What it represents
int_rate	Numeric (continuous)	Interest rate on the loan
loan_amnt	Numeric (continuous)	Amount of the loan issued
annual_inc	Numeric (continuous)	Annual income of the borrower
emp_length	Categorical	Employment length in years (0–10)
dti	Numeric (continuous)	Debt-to-income ratio
grade	Ordinal factor (A–G)	Loan grade assigned by Lending Club; A is best, G is worst
purpose	Nominal factor	Stated purpose of the loan
home_ownership	Nominal factor	Borrower's home ownership status
addr_state	Nominal factor	State of the borrower
term	Ordinal factor	Length of the loan (36 or 60 months)

Data Quality Report

```r
glimpse(actual_data)
```

```
## Rows: 887,379
## Columns: 10
## $ annual_inc     <dbl> 24000.00, 30000.00, 12252.00, 49200.00, 80000.00, 36000~
## $ int_rate       <dbl> 10.65, 15.27, 15.96, 13.49, 12.69, 7.90, 15.96, 18.64, ~
## $ loan_amnt      <dbl> 5000, 2500, 2400, 10000, 3000, 5000, 7000, 3000, 5600, ~
## $ emp_length     <fct> 10+ years, < 1 year, 10+ years, 10+ years, 1 year, 3 ye~
## $ dti            <dbl> 27.65, 1.00, 8.72, 20.00, 17.94, 11.20, 23.51, 5.35, 5.~
## $ addr_state     <fct> AZ, GA, IL, CA, OR, AZ, NC, CA, CA, TX, AZ, CA, VA, IL,~
```

```
## $ grade          <fct> B, C, C, C, B, A, C, E, F, B, C, B, C, B, B, D, C, A, B~
## $ purpose        <fct> credit_card, car, small_business, other, other, wedding~
## $ home_ownership <fct> RENT, RENT, RENT, RENT, RENT, RENT, RENT, RENT, OWN, RE~
## $ term           <fct> 36 months, 60 months, 36 months, 36 months, 60 months, ~
```

```r
dim(actual_data)
```

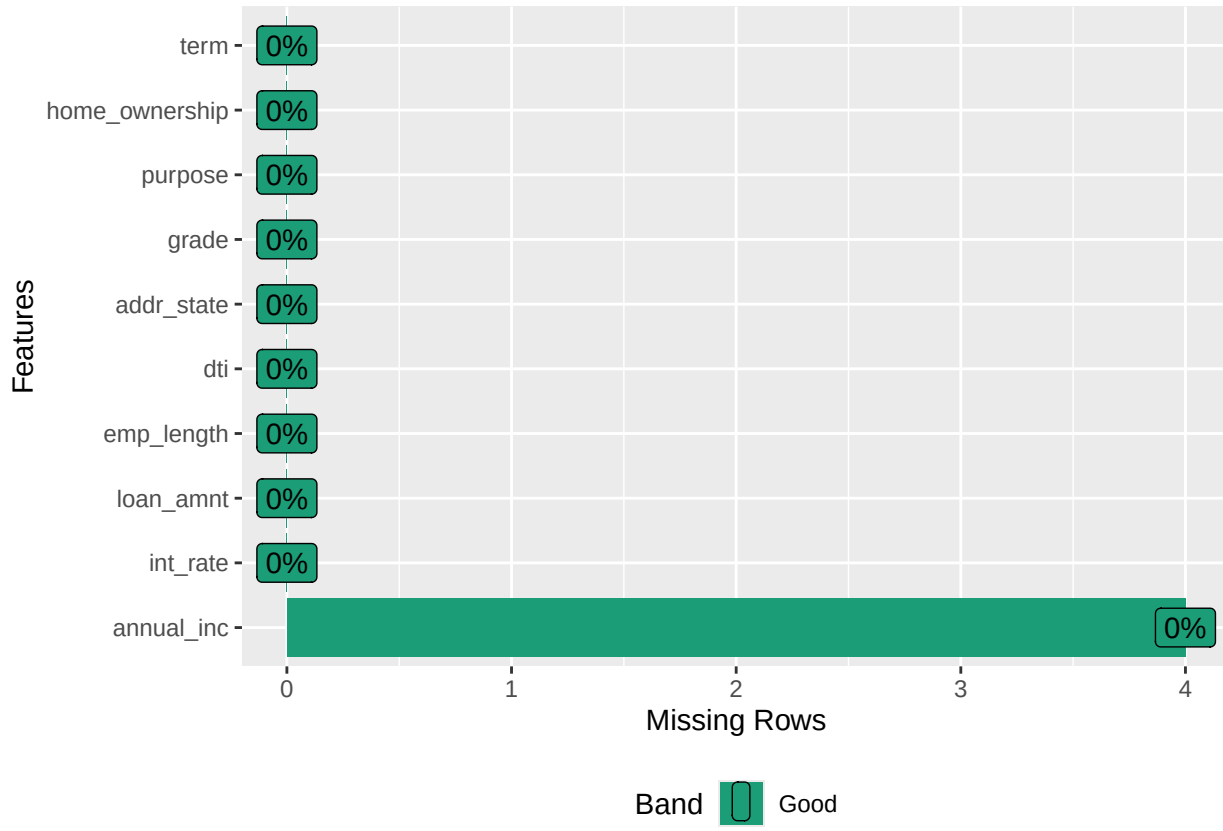
```
## [1] 887379     10
```

```r
DataExplorer::introduce(actual_data)
```

```
## # A tibble: 1 x 9
##     rows columns discrete_columns continuous_columns all_missing_columns
##    <int>   <int>            <int>              <int>               <int>
## 1 887379      10                6                  4                   0
## # i 4 more variables: total_missing_values <int>, complete_rows <int>,
## #   total_observations <int>, memory_usage <dbl>
```

```r
DataExplorer::plot_missing(actual_data)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the DataExplorer package.
##   Please report the issue at
##   <https://github.com/boxuancui/DataExplorer/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
missing_tbl <- actual_data |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Feature", values_to = "n_missing") |>
  arrange(desc(n_missing))

missing_tbl
```

```
## # A tibble: 10 x 2
##    Feature        n_missing
##    <chr>              <int>
##  1 annual_inc             4
##  2 int_rate               0
##  3 loan_amnt              0
##  4 emp_length             0
##  5 dti                    0
##  6 addr_state             0
##  7 grade                  0
##  8 purpose                0
##  9 home_ownership         0
## 10 term                   0
```

```r
any(duplicated(actual_data))
```

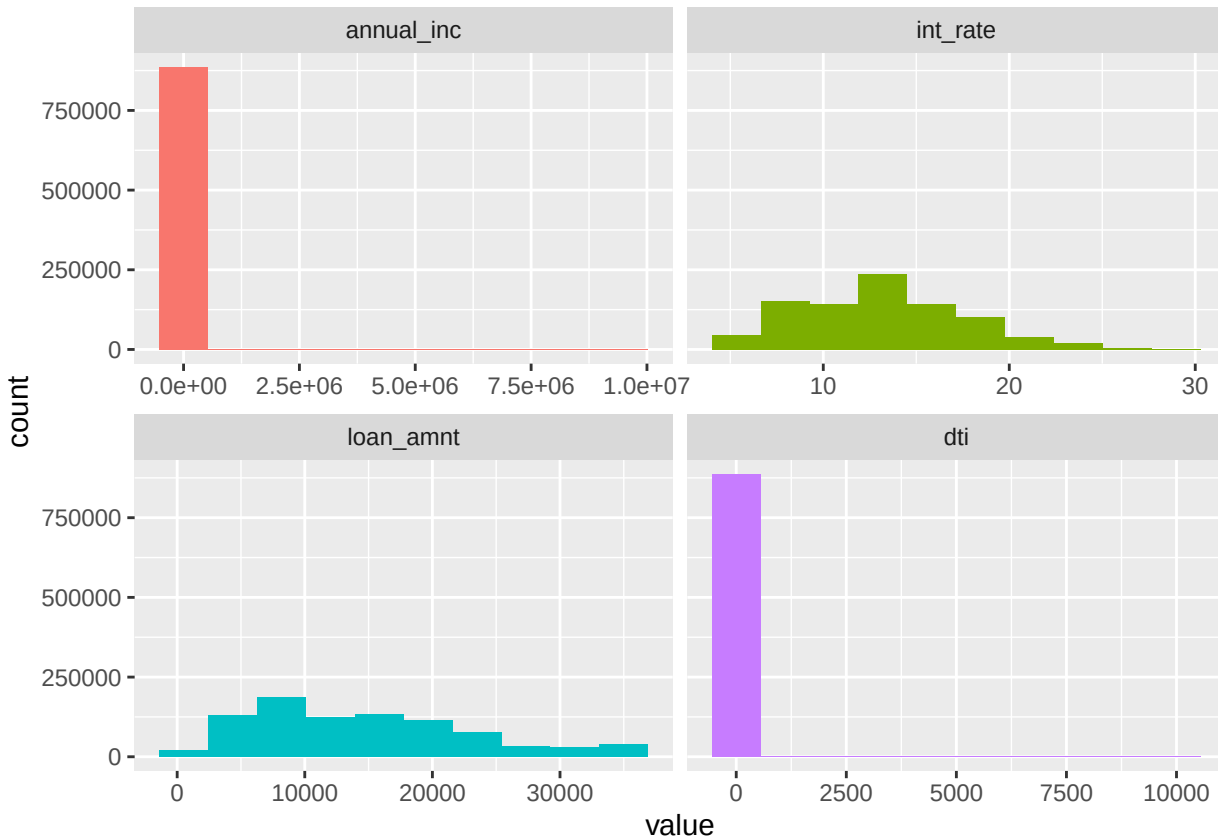
```
## [1] TRUE
```

##Numerical Features

```
numeric_features <- actual_data |>
  select_if(is.numeric)

num_s <- skimr::skim_without_charts(numeric_features)

focus_numeric_table <- as.data.frame(num_s) |>
  mutate(count = nrow(actual_data) - n_missing) |>
  select(skim_variable, count, numeric.p0, numeric.p25, numeric.mean,
         numeric.p50, numeric.p75, numeric.p100, numeric.sd)

extra_two <- funModeling::status(numeric_features)[ ,c("p_na", "unique")]
rownames(extra_two) <- NULL

numeric_report_table <- cbind(focus_numeric_table, extra_two) |>
  relocate(p_na, .before = 3) |>
  relocate(unique, .before = 4)

names(numeric_report_table) <- c("Feature", "Count", "% Miss.", "Cardinality", "Min", "1st Qrt.", "Mean

numeric_report_table
```

```
##      Feature  Count      % Miss. Cardinality    Min 1st Qrt.        Mean
## 1 annual_inc 887375 4.507657e-06       49384   0.00 45000.00 75027.58776
## 2   int_rate 887379 0.000000e+00         542   5.32     9.99    13.24674
## 3  loan_amnt 887379 0.000000e+00        1372 500.00  8000.00 14755.26461
## 4        dti 887379 0.000000e+00        4086   0.00    11.91    18.15704
##     Median 3rd Qrt.       Max    Std. Dev.
## 1 65000.00 90000.00 9.500e+06 64698.300142
## 2    12.99    16.20 2.899e+01     4.381867
## 3 13000.00 20000.00 3.500e+04  8435.455601
## 4    17.65    23.95 9.999e+03    17.190626
```

```
funModeling::plot_num(numeric_features)
```




```r
funModeling::profiling_num(numeric_features)
```

```
##     variable        mean      std_dev variation_coef     p_01     p_05     p_25
## 1 annual_inc 75027.58776 64698.300142      0.8623268 19000.00 28000.00 45000.00
## 2   int_rate    13.24674     4.381867      0.3307884     5.32     6.62     9.99
## 3  loan_amnt 14755.26461  8435.455601      0.5716912  1800.00  3600.00  8000.00
## 4        dti    18.15704    17.190626      0.9467747     1.98     5.20    11.91
##       p_50     p_75      p_95      p_99    skewness     kurtosis      iqr
## 1 65000.00 90000.00 150000.00 250000.00  44.4784117 4.811165e+03 45000.00
## 2    12.99    16.20     20.99     24.99   0.4294799 2.844633e+00     6.21
## 3 13000.00 20000.00  32000.00  35000.00   0.6816820 2.743280e+00 12000.00
## 4    17.65    23.95     32.72     37.48 441.4860288 2.561284e+05    12.04
##          range_98        range_80
## 1 [19000, 250000] [34000, 124800]
## 2   [5.32, 24.99]   [7.69, 18.99]
## 3   [1800, 35000]   [5000, 28000]
## 4   [1.98, 37.48]   [7.45, 29.57]
```

```r
detect_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower <- Q1 - 1.5 * IQR
  upper <- Q3 + 1.5 * IQR
  sum(x < lower | x > upper, na.rm = TRUE)
```

variable	n_outliers
annual_inc	39719
int_rate	6308
loan_amnt	0
dti	81

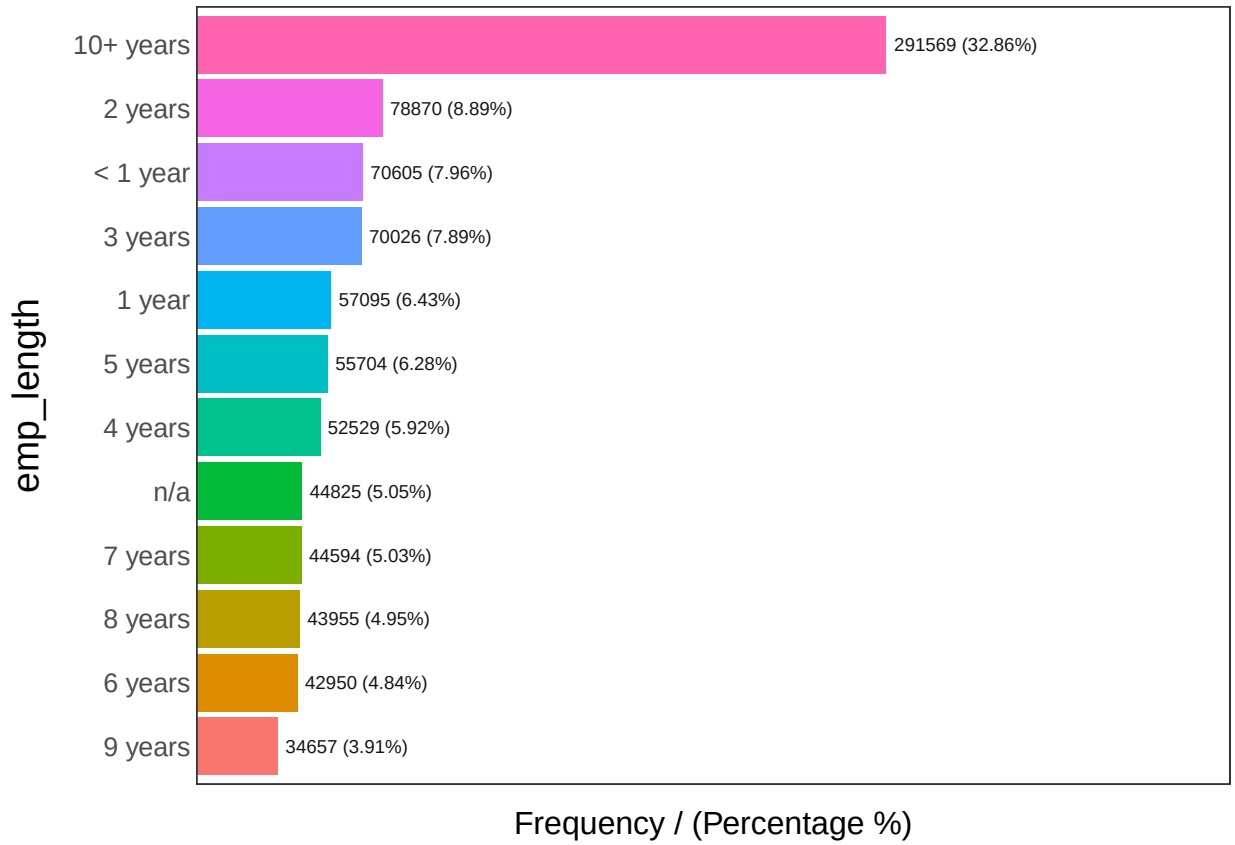
```
}

actual_data |>
  select(annual_inc, int_rate, loan_amnt, dti) |>
  summarise(across(everything(), detect_outliers)) |>
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "n_outliers") |>
  gt()
```

```
categorical_features <- actual_data |>
  select_if(is.factor)

cat_s <- skim_without_charts(categorical_features)

focus_cat_table <- as.data.frame(cat_s) |>
  mutate(count = nrow(actual_data) - n_missing) |>
  mutate(percent_Miss = round((1 - complete_rate) * 100, 1)) |>
  select(skim_variable, count, percent_Miss, factor.n_unique, factor.top_counts)

names(focus_cat_table) <- c("Feature", "Count", "% Miss.", "Card.", "Top Counts")

focus_cat_table
```

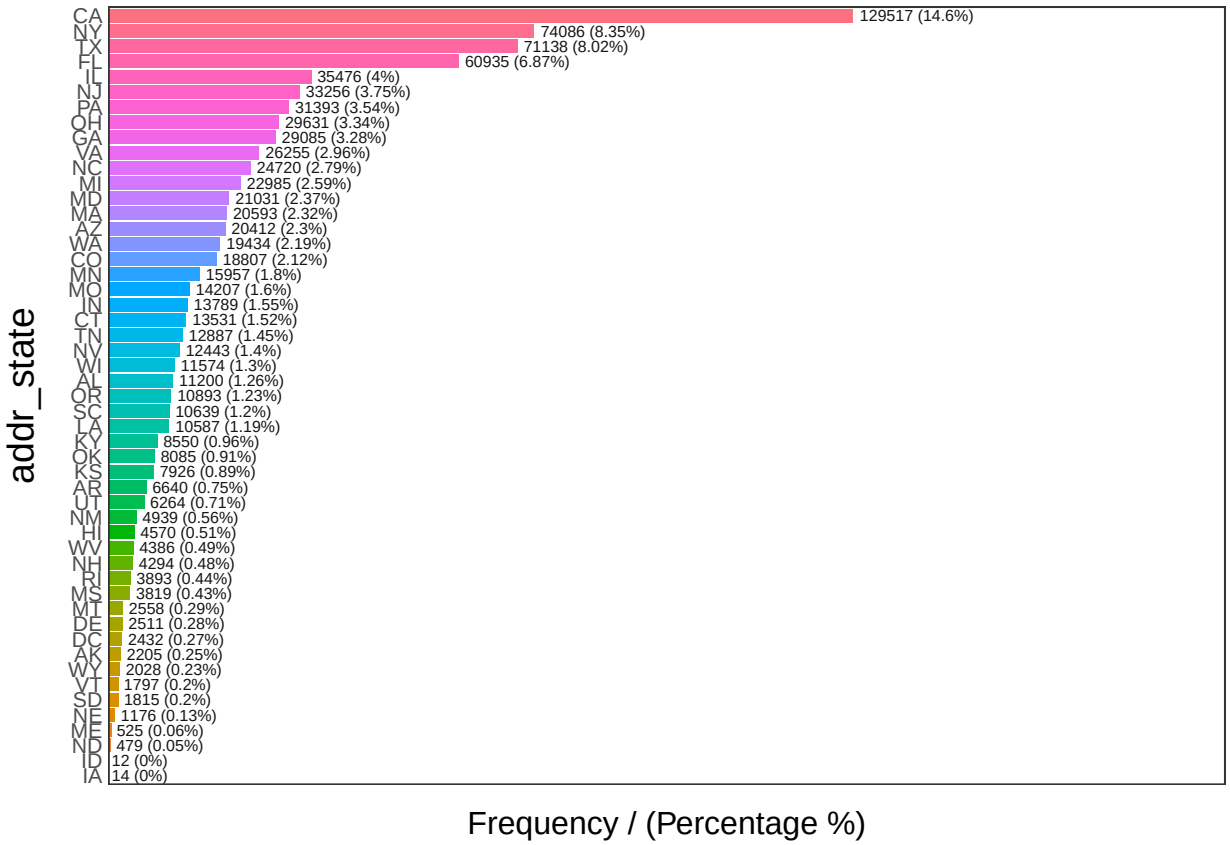
Categorical Features

```
##          Feature  Count % Miss. Card.
## 1     emp_length 887379       0    12
## 2     addr_state 887379       0    51
## 3          grade 887379       0     7
## 4        purpose 887379       0    14
## 5 home_ownership 887379       0     6
## 6           term 887379       0     2
##                                          Top Counts
## 1  10+: 291569, 2 y: 78870, < 1: 70605, 3 y: 70026
## 2      CA: 129517, NY: 74086, TX: 71138, FL: 60935
## 3       B: 254535, C: 245860, A: 148202, D: 139542
## 4 deb: 524215, cre: 206182, hom: 51829, oth: 42894
## 5   MOR: 443557, REN: 356117, OWN: 87470, OTH: 182
## 6                         36 : 621125, 60 : 266254
```

```r
funModeling::freq(categorical_features)
```

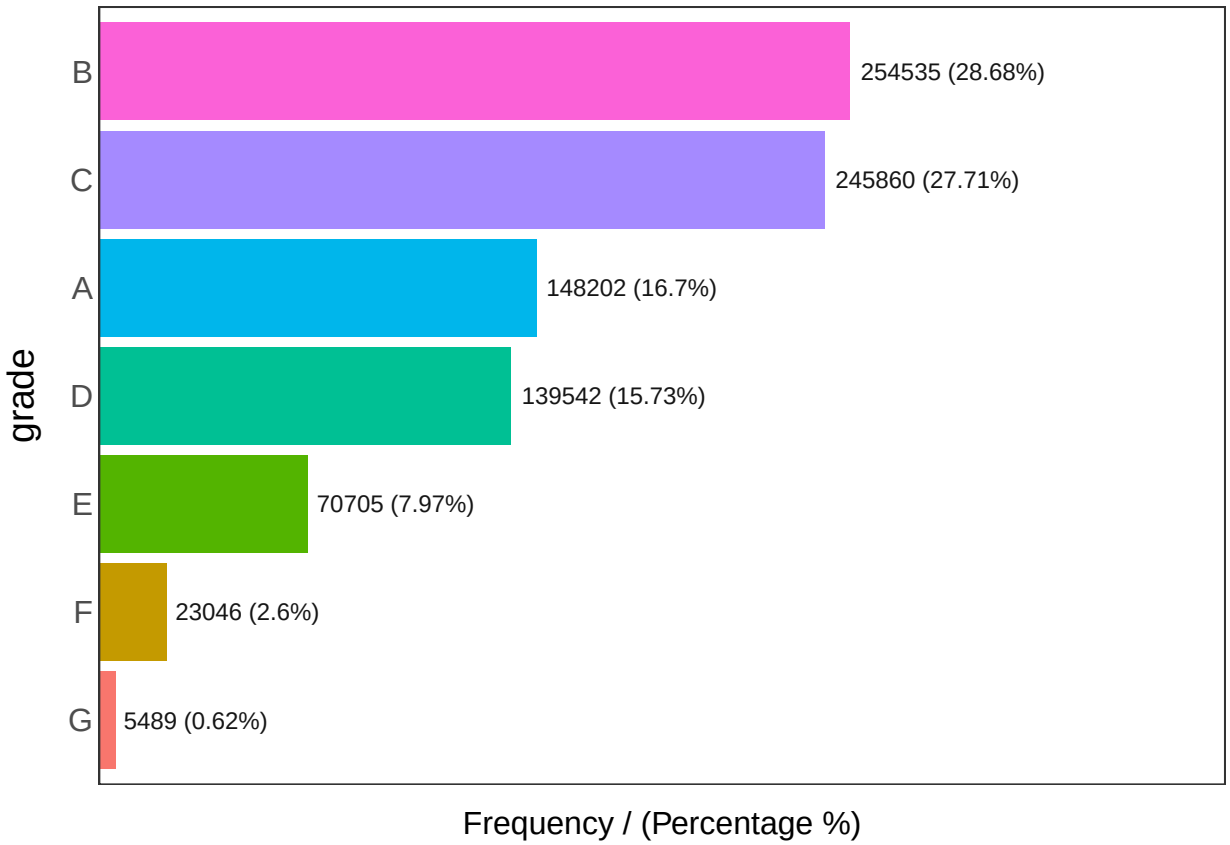



```
##    emp_length frequency percentage cumulative_perc
## 1   10+ years    291569      32.86           32.86
## 2     2 years     78870       8.89           41.75
## 3    < 1 year     70605       7.96           49.71
## 4     3 years     70026       7.89           57.60
## 5      1 year     57095       6.43           64.03
## 6     5 years     55704       6.28           70.31
## 7     4 years     52529       5.92           76.23
## 8         n/a     44825       5.05           81.28
## 9     7 years     44594       5.03           86.31
## 10    8 years     43955       4.95           91.26
## 11    6 years     42950       4.84           96.10
## 12    9 years     34657       3.91          100.00
```

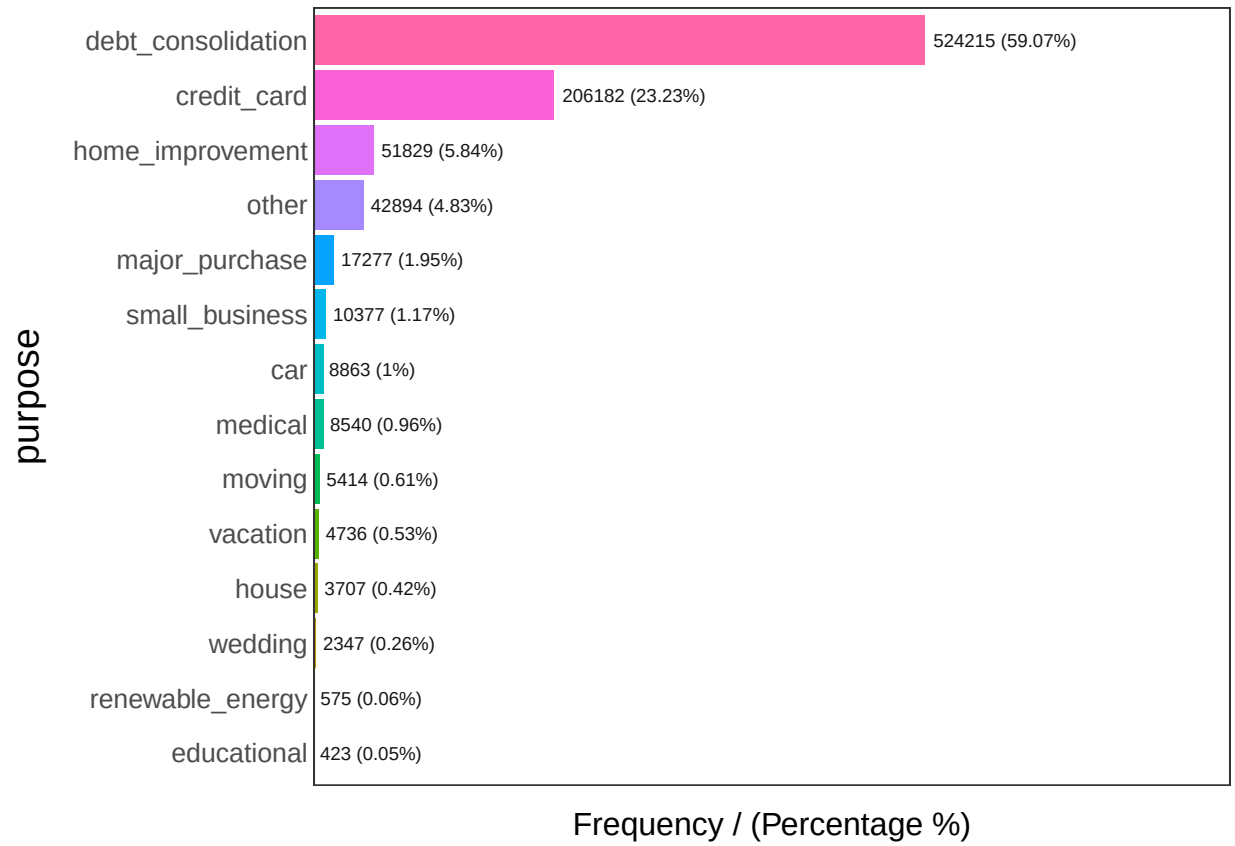



```
##    addr_state frequency percentage cumulative_perc
## 1          CA    129517      14.60           14.60
## 2          NY     74086       8.35           22.95
## 3          TX     71138       8.02           30.97
## 4          FL     60935       6.87           37.84
## 5          IL     35476       4.00           41.84
## 6          NJ     33256       3.75           45.59
## 7          PA     31393       3.54           49.13
## 8          OH     29631       3.34           52.47
## 9          GA     29085       3.28           55.75
## 10         VA     26255       2.96           58.71
## 11         NC     24720       2.79           61.50
## 12         MI     22985       2.59           64.09
## 13         MD     21031       2.37           66.46
## 14         MA     20593       2.32           68.78
## 15         AZ     20412       2.30           71.08
## 16         WA     19434       2.19           73.27
## 17         CO     18807       2.12           75.39
## 18         MN     15957       1.80           77.19
## 19         MO     14207       1.60           78.79
## 20         IN     13789       1.55           80.34
## 21         CT     13531       1.52           81.86
## 22         TN     12887       1.45           83.31
## 23         NV     12443       1.40           84.71
## 24         WI     11574       1.30           86.01
## 25         AL     11200       1.26           87.27
```

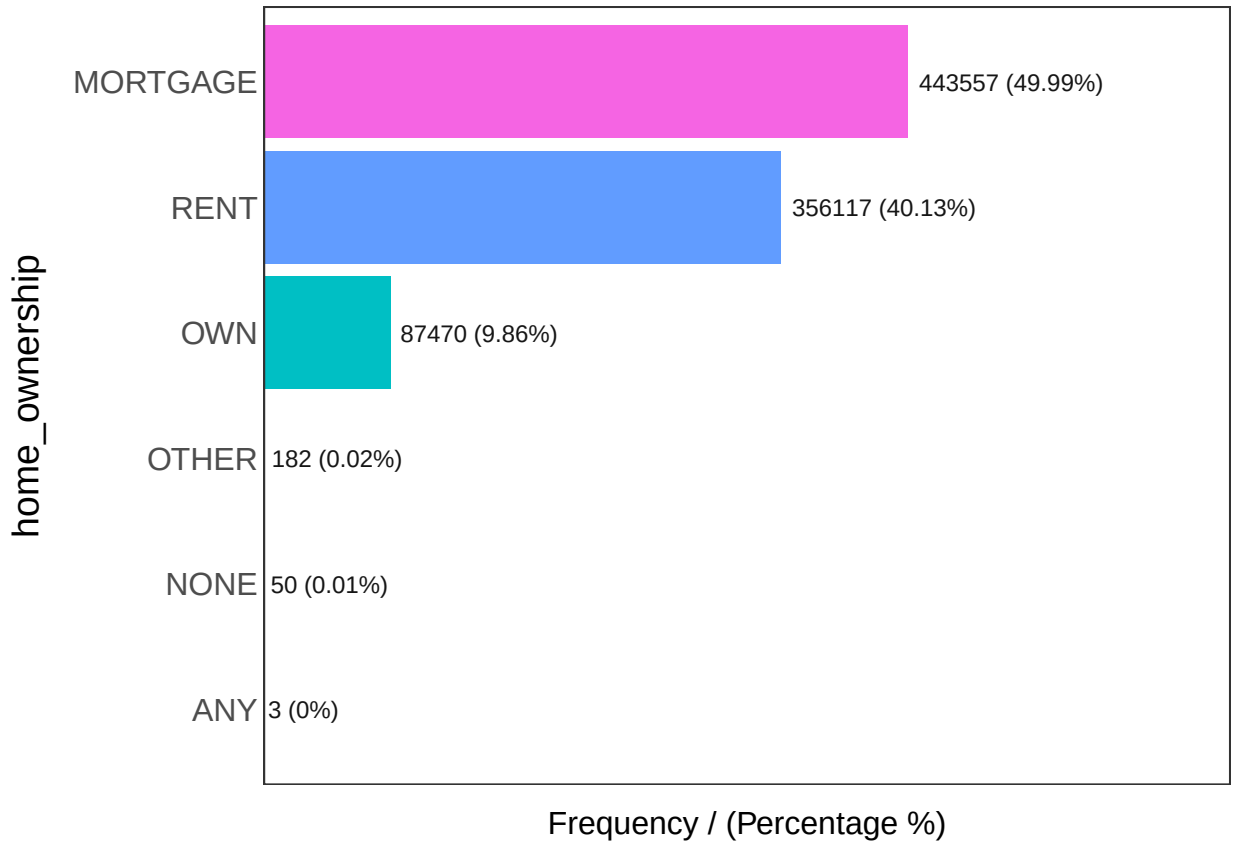
```
## 26          OR      10893        1.23           88.50
## 27          SC      10639        1.20           89.70
## 28          LA      10587        1.19           90.89
## 29          KY       8550        0.96           91.85
## 30          OK       8085        0.91           92.76
## 31          KS       7926        0.89           93.65
## 32          AR       6640        0.75           94.40
## 33          UT       6264        0.71           95.11
## 34          NM       4939        0.56           95.67
## 35          HI       4570        0.51           96.18
## 36          WV       4386        0.49           96.67
## 37          NH       4294        0.48           97.15
## 38          RI       3893        0.44           97.59
## 39          MS       3819        0.43           98.02
## 40          MT       2558        0.29           98.31
## 41          DE       2511        0.28           98.59
## 42          DC       2432        0.27           98.86
## 43          AK       2205        0.25           99.11
## 44          WY       2028        0.23           99.34
## 45          SD       1815        0.20           99.54
## 46          VT       1797        0.20           99.74
## 47          NE       1176        0.13           99.87
## 48          ME        525        0.06           99.93
## 49          ND        479        0.05           99.98
## 50          IA         14        0.00           99.98
## 51          ID         12        0.00          100.00
```



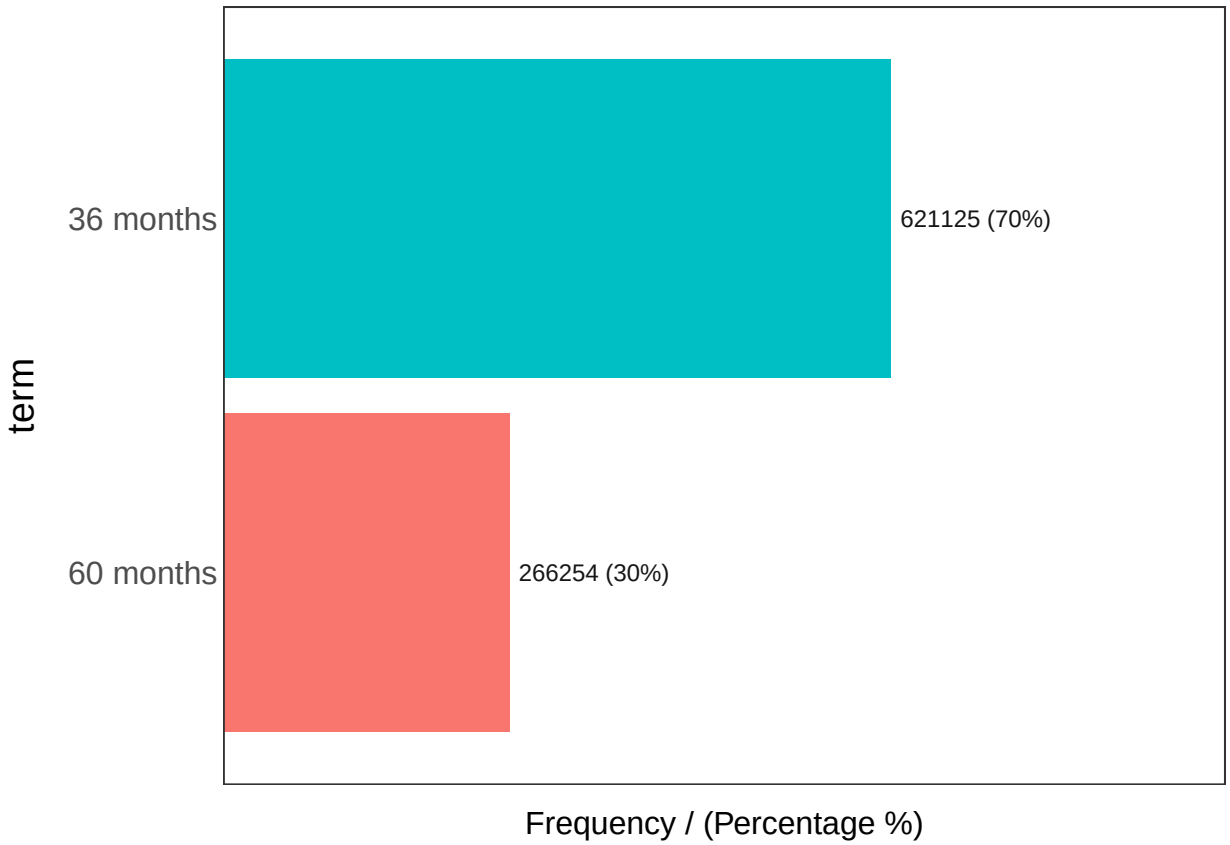
```
##   grade frequency percentage cumulative_perc
## 1     B    254535      28.68           28.68
## 2     C    245860      27.71           56.39
## 3     A    148202      16.70           73.09
## 4     D    139542      15.73           88.82
## 5     E     70705       7.97           96.79
## 6     F     23046       2.60           99.39
## 7     G      5489       0.62          100.00
```




```
##               purpose frequency percentage cumulative_perc
## 1  debt_consolidation    524215      59.07           59.07
## 2         credit_card    206182      23.23           82.30
## 3    home_improvement     51829       5.84           88.14
## 4               other     42894       4.83           92.97
## 5      major_purchase     17277       1.95           94.92
## 6      small_business     10377       1.17           96.09
## 7                 car      8863       1.00           97.09
## 8             medical      8540       0.96           98.05
## 9              moving      5414       0.61           98.66
## 10           vacation      4736       0.53           99.19
## 11              house      3707       0.42           99.61
## 12            wedding      2347       0.26           99.87
## 13   renewable_energy       575       0.06           99.93
## 14        educational       423       0.05          100.00
```




```
##   home_ownership frequency percentage cumulative_perc
## 1       MORTGAGE    443557      49.99           49.99
## 2           RENT    356117      40.13           90.12
## 3            OWN     87470       9.86           99.98
## 4          OTHER       182       0.02          100.00
## 5           NONE        50       0.01          100.01
## 6            ANY         3       0.00          100.00
```




```
##        term frequency percentage cumulative_perc
## 1 36 months    621125         70              70
## 2 60 months    266254         30             100
```

```
## [1] "Variables processed: emp_length, addr_state, grade, purpose, home_ownership, term"
```

Data Issues and Proposed Handling Techniques

The following issues were identified in the raw dataset and addressed during data cleaning:

Feature	Data Quality Issue	Proposed Handling Solution
annual_inc	4 missing values (0.0005%)	Retain in dataset; drop only within chunks that require it
annual_inc	39,719 outliers via IQR; max $9,500,000; skewness 44.48, kurtosis 4,811 — severe right skew	Valid outliers (high earners exist); apply clamp transformation (Q1 - 1.5×IQR, Q3 + 1.5×IQR)
int_rate	6,308 outliers via IQR; range 5.32–28.99% realistic; slight right skew (0.43)	Valid outliers (higher rates reflect riskier borrowers); apply clamp transformation
loan_amnt	0 missing values; 0 outliers; range $500–$35,000 within LendingClub limits	No action needed

Feature	Data Quality Issue	Proposed Handling Solution
dti	81 outliers via IQR; max of 9,999 is almost certainly invalid; skewness 441, kurtosis 256,128	Invalid outlier; remove rows where dti > 100
emp_length	`n/a` present as a factor level (44,825 rows / 5.05%) — hidden missing values	Recode `n/a` to `NA`; retain rows for analyses not requiring emp_length
addr_state	Cardinality of 51 — expected (50 states + DC); no unexpected levels	No action needed
grade	Cardinality of 7 (A–G); skewed toward B (28.68%) and C (27.71%)	Valid distribution; encode as ordered factor A→G
purpose	Cardinality of 14; rare categories renewable_energy (0.06%) and educational (0.05%)	Consider collapsing rare categories into `other` for cleaner analysis
home_ownership	OTHER (182, 0.02%), NONE (50, 0.01%), ANY (3, 0%) represent <0.03% combined	Collapse OTHER, NONE, ANY into single OTHER category
term	Cardinality of 2; 70/30 split between 36 and 60 months — imbalanced but realistic	Valid distribution; encode as ordered factor
All character columns	Potential inconsistent spacing and blank-like values (e.g. `n/a` in emp_length)	Apply str_squish(), na_if(), and case_when() standardization

Data Cleaning

```r
# Step 1: Recode n/a in emp_length to NA and clean blank-like string values
actual_data <- actual_data |>
  mutate(across(where(is.factor), ~ fct_recode(., NULL = "n/a"))) |>
  mutate(across(where(is.character), ~ str_squish(.))) |>
  mutate(across(where(is.character), ~ na_if(.x, ""))) |>
  mutate(across(where(is.character),
    ~ case_when(
      .x %in% c("NA", "N/A", "null", "NULL", "?", "unknown", "Unknown") ~ NA_character_,
      TRUE ~ .x
    )))

# Step 2: Remove duplicate rows (keep first occurrence)
actual_data <- actual_data[!duplicated(actual_data), ]

# Step 3: Remove invalid dti outliers
actual_data <- actual_data |>
  filter(dti <= 100)

# Step 4: Collapse rare home_ownership levels into OTHER
actual_data <- actual_data |>
  mutate(home_ownership = fct_collapse(home_ownership,
    OTHER = c("OTHER", "NONE", "ANY")
  ))
```

```r
# Step 5: Ordered factor encoding
actual_data <- actual_data |>
  mutate(
    grade = factor(grade,
                   levels = c("A","B","C","D","E","F","G"),
                   ordered = TRUE),
    term = factor(term,
                  levels = c("36 months","60 months"),
                  ordered = TRUE),
    emp_length = factor(emp_length,
                   levels = c("< 1 year", "1 year", "2 years", "3 years",
                              "4 years", "5 years", "6 years", "7 years",
                              "8 years", "9 years", "10+ years"),
                   ordered = TRUE),
    purpose = fct_relabel(purpose, ~ str_replace_all(.x, "_", " "))
  )

# Step 6: Apply clamp transformation to valid outliers
clamp <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  lower <- Q1 - 1.5 * IQR_val
  upper <- Q3 + 1.5 * IQR_val
  pmax(pmin(x, upper), lower)
}

actual_data <- actual_data |>
  mutate(
    annual_inc = clamp(annual_inc),
    int_rate   = clamp(int_rate)
  )

# Step 7: Drop rows with NAs in key variables
# (retain rows missing only emp_length or annual_inc for unrelated analyses)
actual_data <- actual_data |>
  drop_na(int_rate, loan_amnt, grade, home_ownership,
          purpose, term, addr_state, dti)

# Step 8: Confirm final state of data
glimpse(actual_data)
```

```
## Rows: 887,365
## Columns: 10
## $ annual_inc     <dbl> 24000.00, 30000.00, 12252.00, 49200.00, 80000.00, 36000~
## $ int_rate       <dbl> 10.65, 15.27, 15.96, 13.49, 12.69, 7.90, 15.96, 18.64, ~
## $ loan_amnt      <dbl> 5000, 2500, 2400, 10000, 3000, 5000, 7000, 3000, 5600, ~
## $ emp_length     <ord> 10+ years, < 1 year, 10+ years, 10+ years, 1 year, 3 ye~
## $ dti            <dbl> 27.65, 1.00, 8.72, 20.00, 17.94, 11.20, 23.51, 5.35, 5.~
## $ addr_state     <fct> AZ, GA, IL, CA, OR, AZ, NC, CA, CA, TX, AZ, CA, VA, IL,~
## $ grade          <ord> B, C, C, C, B, A, C, E, F, B, C, B, C, B, B, D, C, A, B~
## $ purpose        <fct> credit card, car, small business, other, other, wedding~
## $ home_ownership <fct> RENT, RENT, RENT, RENT, RENT, RENT, RENT, RENT, OWN, RE~
```

```
## $ term           <ord> 36 months, 60 months, 36 months, 36 months, 60 months, ~
```

```r
dim(actual_data)
```

```
## [1] 887365     10
```

```r
(missing_tbl_final <- actual_data |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Feature", values_to = "n_missing") |>
  arrange(desc(n_missing)))
```

```
## # A tibble: 10 x 2
##    Feature        n_missing
##    <chr>              <int>
##  1 emp_length         44818
##  2 annual_inc             4
##  3 int_rate               0
##  4 loan_amnt              0
##  5 dti                    0
##  6 addr_state             0
##  7 grade                  0
##  8 purpose                0
##  9 home_ownership         0
## 10 term                   0
```

Creator: Tracy Huang

Q1: Can interest rate by predicted based on loan term and loan grade?

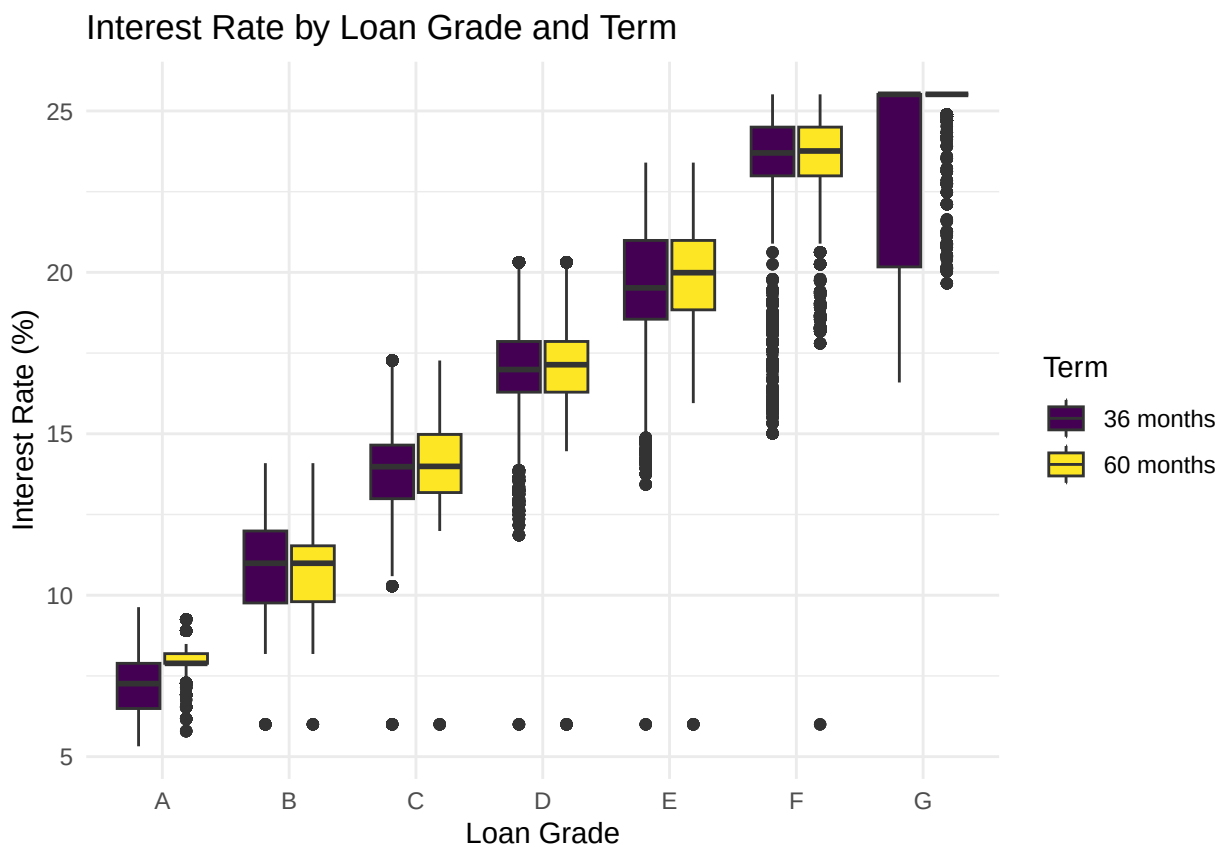
Summary:

```r
q1_table <- actual_data |>
  group_by(grade, term) |>
  summarise(
    n = n(),
    mean_int_rate = round(mean(int_rate, na.rm = TRUE), 2),
    median_int_rate = round(median(int_rate, na.rm = TRUE), 2),
    sd_int_rate = round(sd(int_rate, na.rm = TRUE), 2),
    q1 = round(quantile(int_rate, 0.25, na.rm = TRUE), 2),
    q3 = round(quantile(int_rate, 0.75, na.rm = TRUE), 2),
    .groups = "drop"
  ) |>
  arrange(grade, term)

q1_table |> gt()
```

Graph:

grade	term	n	mean_int_rate	median_int_rate	sd_int_rate	q1	q3
A	36 months	143015	7.22	7.26	0.98	6.49	7.89
A	60 months	5187	8.00	7.89	0.54	7.89	8.19
B	36 months	212007	10.86	10.99	1.46	9.76	11.99
B	60 months	42525	10.66	10.99	1.35	9.80	11.53
C	36 months	162418	13.92	13.98	1.22	12.99	14.65
C	60 months	83440	14.09	13.99	1.25	13.18	14.98
D	36 months	76683	17.16	16.99	1.29	16.29	17.86
D	60 months	62853	17.19	17.14	1.18	16.29	17.86
E	36 months	21841	19.76	19.52	1.64	18.55	20.99
E	60 months	48861	19.96	19.99	1.43	18.84	20.99
F	36 months	4466	23.32	23.70	2.05	22.99	24.50
F	60 months	18580	23.63	23.76	1.35	22.99	24.50
G	36 months	686	23.40	25.51	3.12	20.17	25.51
G	60 months	4803	25.18	25.51	1.02	25.51	25.51

```r
actual_data |>
  ggplot(aes(x = grade, y = int_rate, fill = term)) +
  geom_boxplot() +
  labs(
    title = "Interest Rate by Loan Grade and Term",
    x = "Loan Grade",
    y = "Interest Rate (%)",
    fill = "Term"
  ) +
  theme_minimal()
```



Interpretation of Findings: Interest rate is almost perfectly explained by grade and term together. Grade A loans average 7.22–8.00%, while grade G loans average 23.40–25.18%, a spread of over 17 percentage points. Within each grade, 60-month loans consistently carry slightly higher rates than 36-month loans, though the gap is modest (under 1%). The monotonic increase from A through G confirms that LendingClub's internal grading system is highly consistent with its pricing engine. For risk assessment purposes, grade is the dominant pricing signal, and loan term acts as a secondary risk premium on top of it.

Q2: Can home ownership status be predicted using borrower income and employment length?

Summary:

```
q2_table <- actual_data |>
  mutate(inc_decile = ntile(annual_inc, 10)) |>
  group_by(home_ownership, inc_decile) |>
  summarise(
    n = n(),
    inc_min = round(min(annual_inc, na.rm = TRUE), 0),
    inc_max = round(max(annual_inc, na.rm = TRUE), 0),
    mean_inc = round(mean(annual_inc, na.rm = TRUE), 0),
    .groups = "drop"
  ) |>
  arrange(home_ownership, inc_decile)
```

```
## Warning: There were 2 warnings in `summarise()`.
## The first warning was:
```

```
## i In argument: `inc_min = round(min(annual_inc, na.rm = TRUE), 0)`.
## i In group 11: `home_ownership = OTHER` `inc_decile = NA`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

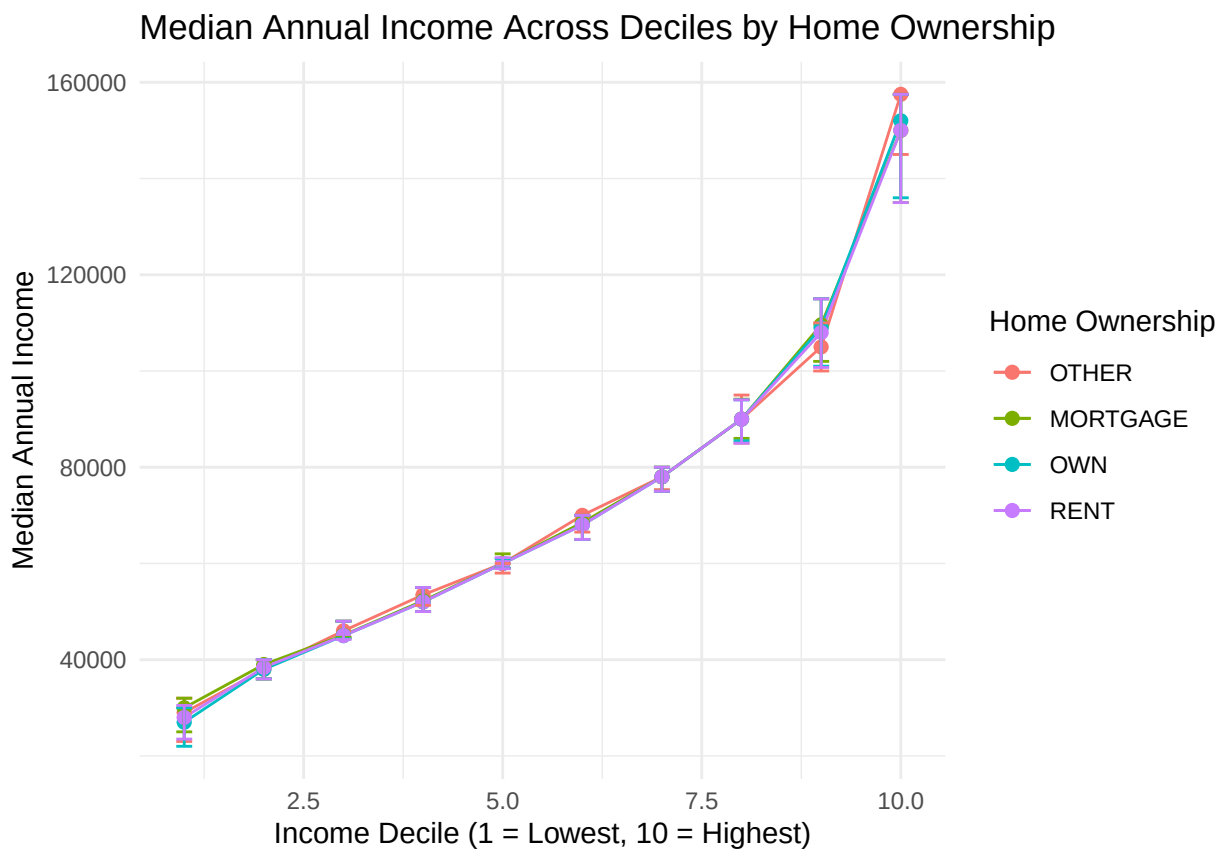
```r
q2_table |> gt()
```

Graph:

```r
q2_plot_data <- actual_data |>
  mutate(inc_decile = ntile(annual_inc, 10)) |>
  group_by(home_ownership, inc_decile) |>
  summarise(
    median_inc = median(annual_inc, na.rm = TRUE),
    q1 = quantile(annual_inc, 0.25, na.rm = TRUE),
    q3 = quantile(annual_inc, 0.75, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(q2_plot_data,
       aes(x = inc_decile, y = median_inc,
           group = home_ownership, color = home_ownership)) +
  geom_line() +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = q1, ymax = q3), width = 0.2) +
  labs(
    title = "Median Annual Income Across Deciles by Home Ownership",
    x = "Income Decile (1 = Lowest, 10 = Highest)",
    y = "Median Annual Income",
    color = "Home Ownership"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Interpretation of Findings: Income alone does a poor job of separating ownership groups, all four categories (RENT, MORTGAGE, OWN, OTHER) track nearly identical income trajectories across all 10 deciles, with medians converging tightly from roughly $27k at decile 1 to $147k at decile 10. This means income does not cleanly predict whether a borrower rents or owns. However, the homeownership breakdown by employment length (in Emily's Q2 follow-up) adds important context: as employment length increases, the share of mortgage holders rises steadily from 35% (under 1 year) to 62% (10+ years), while renters fall from 56% to 28%. Together, this suggests employment stability is a better differentiator of housing status than income level alone, and that renters — who are often earlier in their careers may represent a higher-mobility, higher-turnover risk segment for LendingClub.

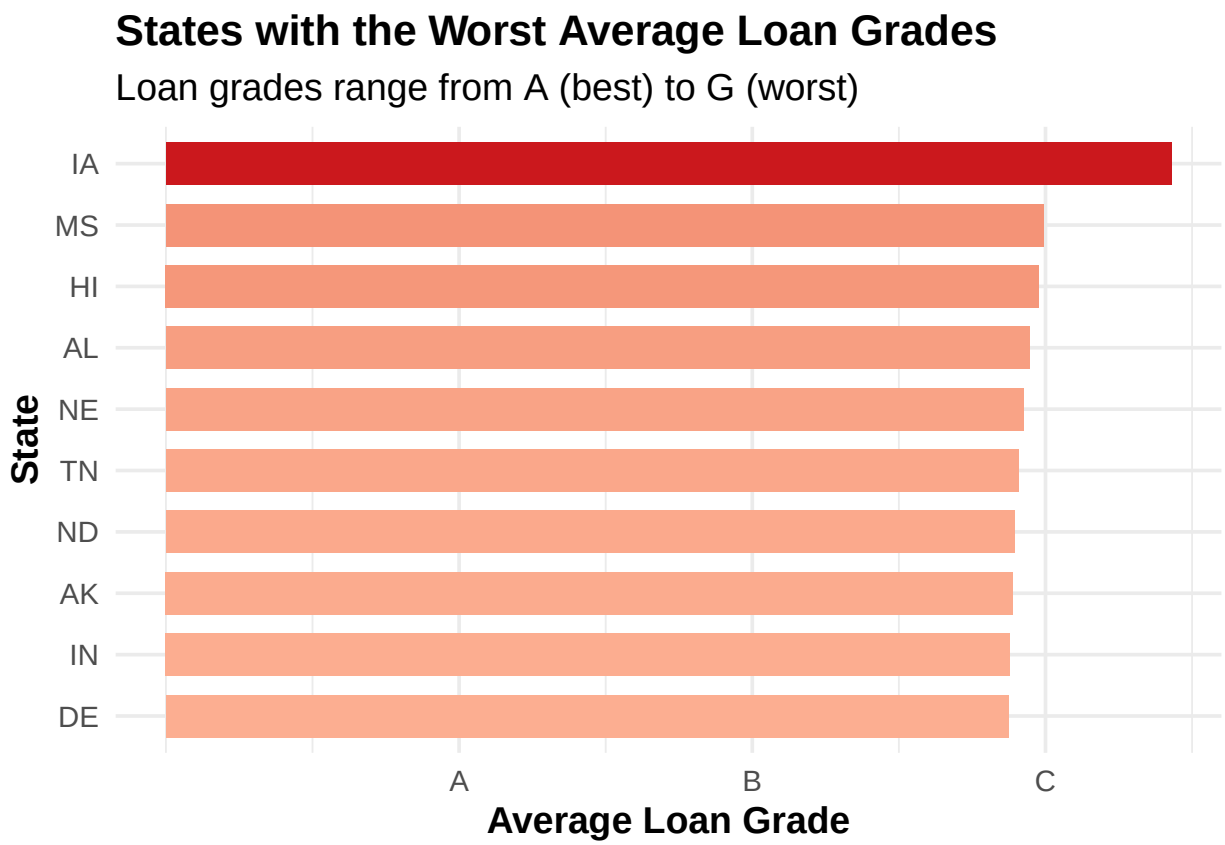
Interpreter: Kyran Lakhia

Q1: Which states are experiencing the lowest grades on average?

```
# Summary table: average grade by state (bottom 10)
state_avg <- actual_data |>
  mutate(grade_num = as.numeric(grade)) |>
  group_by(addr_state) |>
  summarise(avg_grade = mean(grade_num, na.rm = TRUE)) |>
  arrange(desc(avg_grade)) |>
  slice_head(n = 10) |>
  mutate(avg_grade_label = c("A","B","C","D","E","F","G")[round(avg_grade)])

state_avg
```

```
## # A tibble: 10 x 3
##    addr_state avg_grade avg_grade_label
##    <fct>          <dbl> <chr>
##  1 IA              3.43 C
##  2 MS              2.99 C
##  3 HI              2.98 C
##  4 AL              2.95 C
##  5 NE              2.92 C
##  6 TN              2.91 C
##  7 ND              2.90 C
##  8 AK              2.89 C
##  9 IN              2.88 C
## 10 DE              2.87 C
```

```r
# Bar chart: states with worst average loan grades
ggplot(state_avg, aes(x = reorder(addr_state, avg_grade), y = avg_grade, fill = avg_grade)) +
  geom_col(width = 0.7) +
  coord_flip() +
  scale_fill_gradient(low = "#fcae91", high = "#cb181d") +
  scale_y_continuous(
    breaks = 1:7,
    labels = c("A","B","C","D","E","F","G")
  ) +
  labs(
    title = "States with the Worst Average Loan Grades",
    subtitle = "Loan grades range from A (best) to G (worst)",
    x = "State",
    y = "Average Loan Grade"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 16),
    axis.title = element_text(face = "bold")
  )
```



Interpretation: No state is dramatically underperforming. The ten worst-graded states (Iowa, Mississippi, Hawaii, Alabama, Nebraska, Tennessee, North Dakota, Alaska, Indiana, Delaware) all average between grade B and C numerically. Iowa leads with the highest average numeric grade at 3.43, still mapping to a "C." This is actually a reassuring signal: even in the weakest states, credit quality remains in acceptable territory. However, for LendingClub's risk team, these states merit closer monitoring. Smaller-volume states like Iowa (only 14 loans) and Idaho (12 loans) should be interpreted with caution, as their averages are based on tiny samples and could shift significantly with more data.

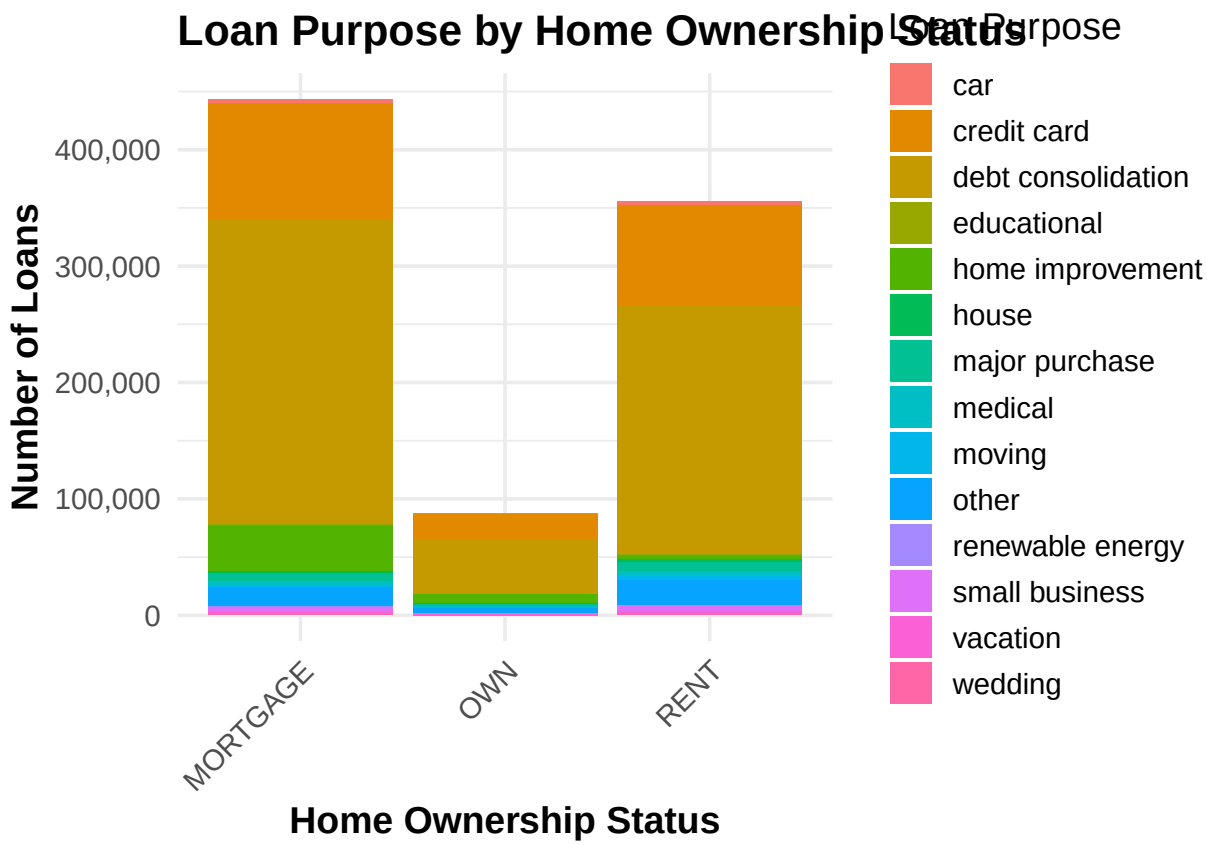
Q2: Is there a relationship between home ownership and purpose?

```r
# Summary table: contingency table of home ownership x purpose
table(actual_data$home_ownership, actual_data$purpose)
```

```
##
##                car credit card debt consolidation educational home improvement
##   OTHER          3          71                 78           5               15
##   MORTGAGE    3660      100160             261859         113            40040
##   OWN         1098       20751              47562          34             7802
##   RENT        4102       85198             214706         271             3971
##
##              house major purchase medical moving  other renewable energy
##   OTHER          3              8       5      3     28                1
##   MORTGAGE    1240           7292    3830    915  16934              236
##   OWN          466           2015     921    301   4772               73
```

```
##   RENT       1998           7962    3784   4195  21159              265
##
##            small business vacation wedding
##   OTHER                14        0       1
##   MORTGAGE           4829     1697     743
##   OWN                 994      530     151
##   RENT               4540     2509    1452
```

```r
# Stacked bar chart: loan purpose by home ownership status
actual_data |>
  filter(home_ownership %in% c("RENT", "MORTGAGE", "OWN")) |>
  group_by(home_ownership, purpose) |>
  summarise(count = n(), .groups = "drop") |>
  ggplot(aes(x = home_ownership, y = count, fill = purpose)) +
  geom_bar(stat = "identity") +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Loan Purpose by Home Ownership Status",
    x = "Home Ownership Status",
    y = "Number of Loans",
    fill = "Loan Purpose"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.title = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



Interpretation: Across all three major ownership groups, debt consolidation dominates loan purpose, accounting for roughly 60% of loans regardless of whether the borrower rents, has a mortgage, or owns outright. Credit cards are the second most common purpose across all groups. One notable difference: renters show a disproportionately low share of home improvement loans compared to mortgage holders, which is intuitive since renters have less incentive to invest in property upgrades. The near-uniform distribution of purposes across ownership types suggests that home ownership is not a strong differentiator of why someone borrows, but it may still be meaningful for predicting repayment behavior, since mortgage holders tend to have more financial obligations and stability.

Orator 1: Ross Guinan

Q1: Is there a relationship between loan amount and loan grade?

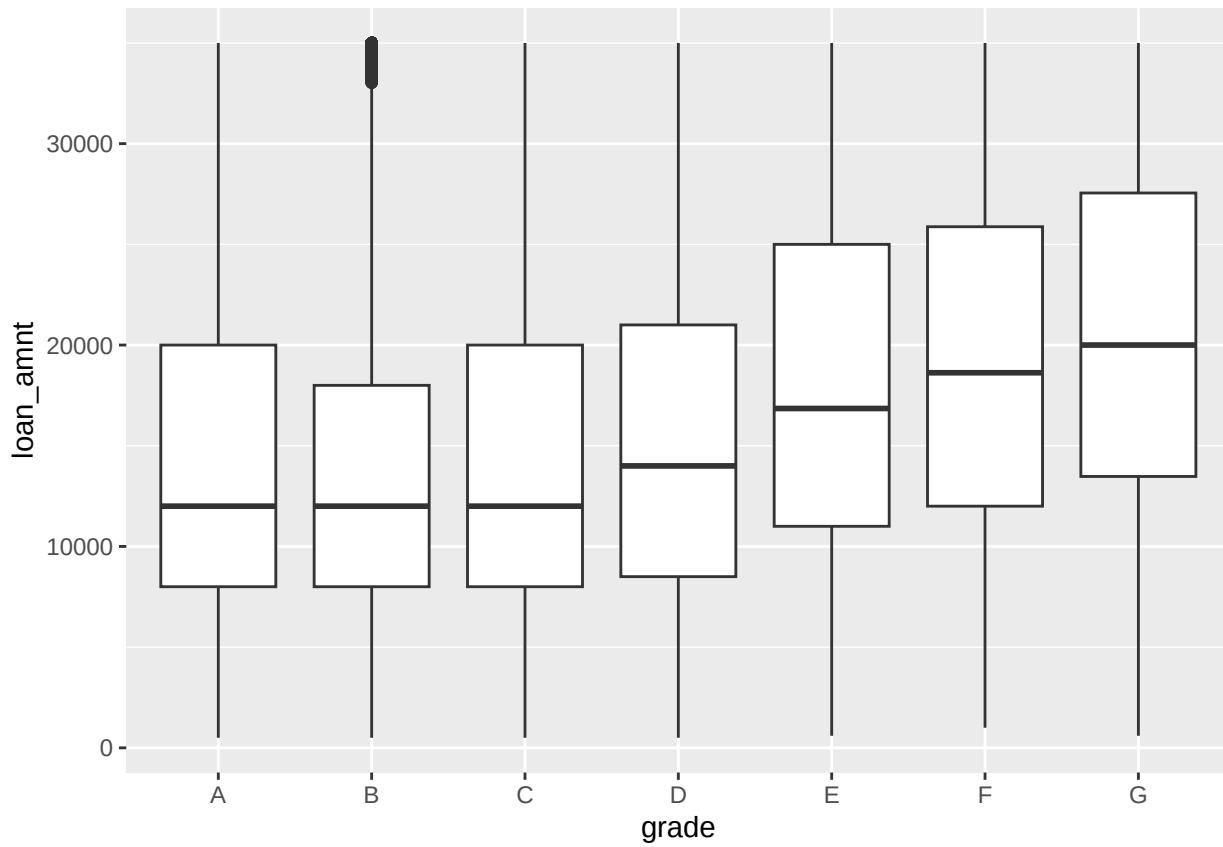
Summary:

```
grade_table <- actual_data |>
  group_by(grade) |>
  summarise(
    n = n(),
    mean_loan_amnt = round(mean(loan_amnt, na.rm = TRUE), 2),
    median_loan_amnt = round(median(loan_amnt, na.rm = TRUE), 2),
    sd_loan_amnt = round(sd(loan_amnt, na.rm = TRUE), 2),
    q1 = round(quantile(loan_amnt, 0.25, na.rm = TRUE), 2),
    q3 = round(quantile(loan_amnt, 0.75, na.rm = TRUE), 2),
    .groups = "drop"
```

```
  ) |>
  arrange(grade)

grade_table |> gt()
```

Graph:

```
q1_boxplot <- ggplot(actual_data, aes(x = grade, y = loan_amnt)) +
  geom_boxplot()

q1_boxplot
```



Interpretation of Findings: There is a clear, positive relationship between loan grade and loan amount — but it runs in the counterintuitive direction. Grade A borrowers (the most creditworthy) have a median loan of $12,000, while grade G borrowers (the riskiest) have a median of $20,000. This means riskier borrowers are taking out larger loans, which compounds LendingClub's exposure. The boxplots also show increasing spread and higher upper quartiles as grade worsens, confirming that high-grade borrowers tend to take smaller, more standardized loans while low-grade borrowers show more variability. From a portfolio management perspective, the combination of lower creditworthiness and larger loan sizes in grades E–G represents a concentrated risk that warrants careful underwriting scrutiny.

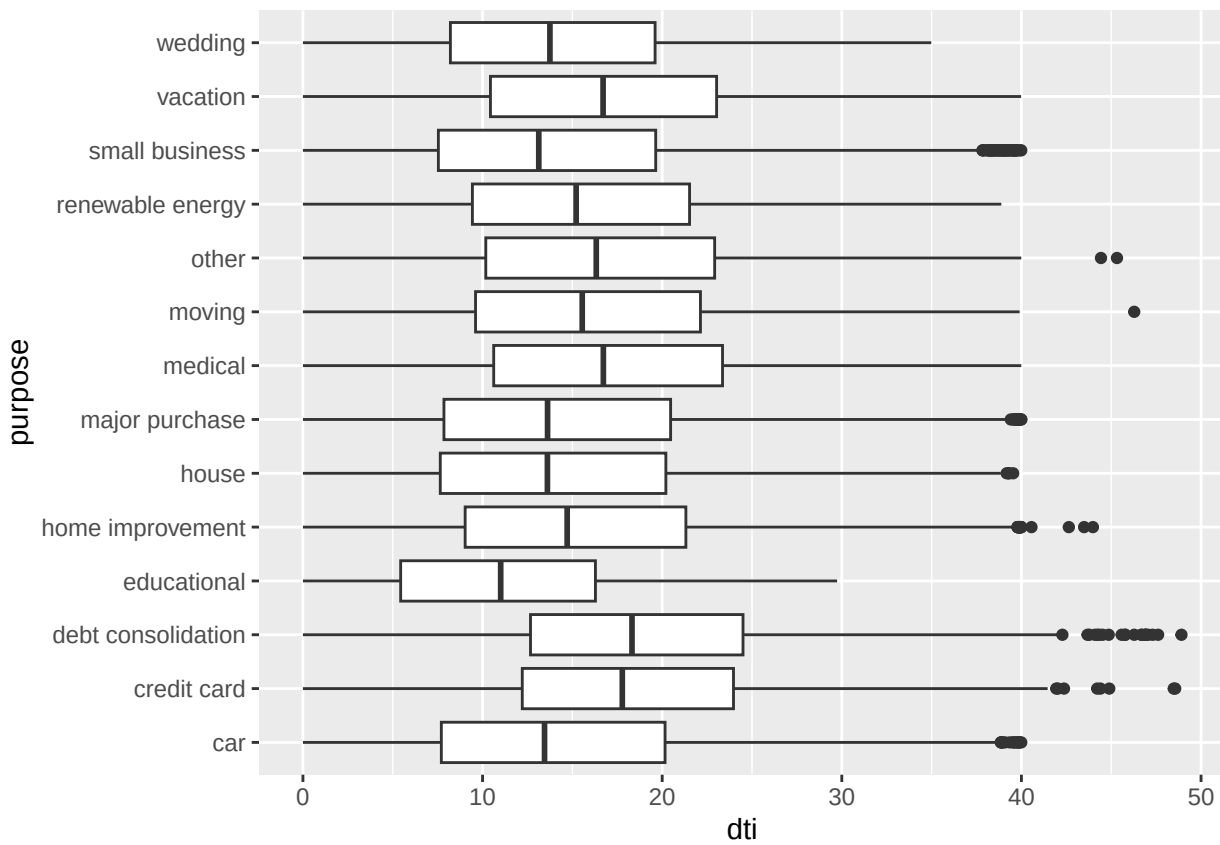
Q2: Is there a correlation between DTI (debt-to-income) and purpose?

Summary:

```r
ross_q2_table <- actual_data |>
  group_by(purpose) |>
  summarise(
    n = n(),
    mean_dti = round(mean(dti, na.rm = TRUE), 2),
    median_dti = round(median(dti, na.rm = TRUE), 2),
    sd_dti = round(sd(dti, na.rm = TRUE), 2),
    q1 = round(quantile(dti, 0.25, na.rm = TRUE), 2),
    q3 = round(quantile(dti, 0.75, na.rm = TRUE), 2),
    .groups = "drop"
  ) |>
  arrange(purpose)

ross_q2_table |> gt()
```

Graph:

```r
capped_dti <- actual_data[actual_data$dti <= 50,]
q2_boxplot <- ggplot(capped_dti, aes(x = purpose, y = dti)) +
  geom_boxplot() +
  coord_flip()


q2_boxplot
```



Interpretation of Findings: Debt consolidation and credit card loans — which together make up over 82% of all loans — show the highest median DTI values, around 18–19. This makes intuitive sense: borrowers already carrying heavy debt loads are turning to LendingClub to restructure or pay off existing obligations. In contrast, car loans, small business loans, and educational loans show notably lower DTI medians (around 11–14), suggesting these borrowers come in with lighter existing debt burdens. High-DTI borrowers concentrated in debt consolidation are a particular risk flag and they are already financially stretched, and if the consolidation loan fails to meaningfully reduce their obligations, the probability of default rises. LendingClub should weight DTI heavily when evaluating consolidation loan applications.

Orator 2: Carter Boxell

Q1: How does the average interest rates for each grade compare

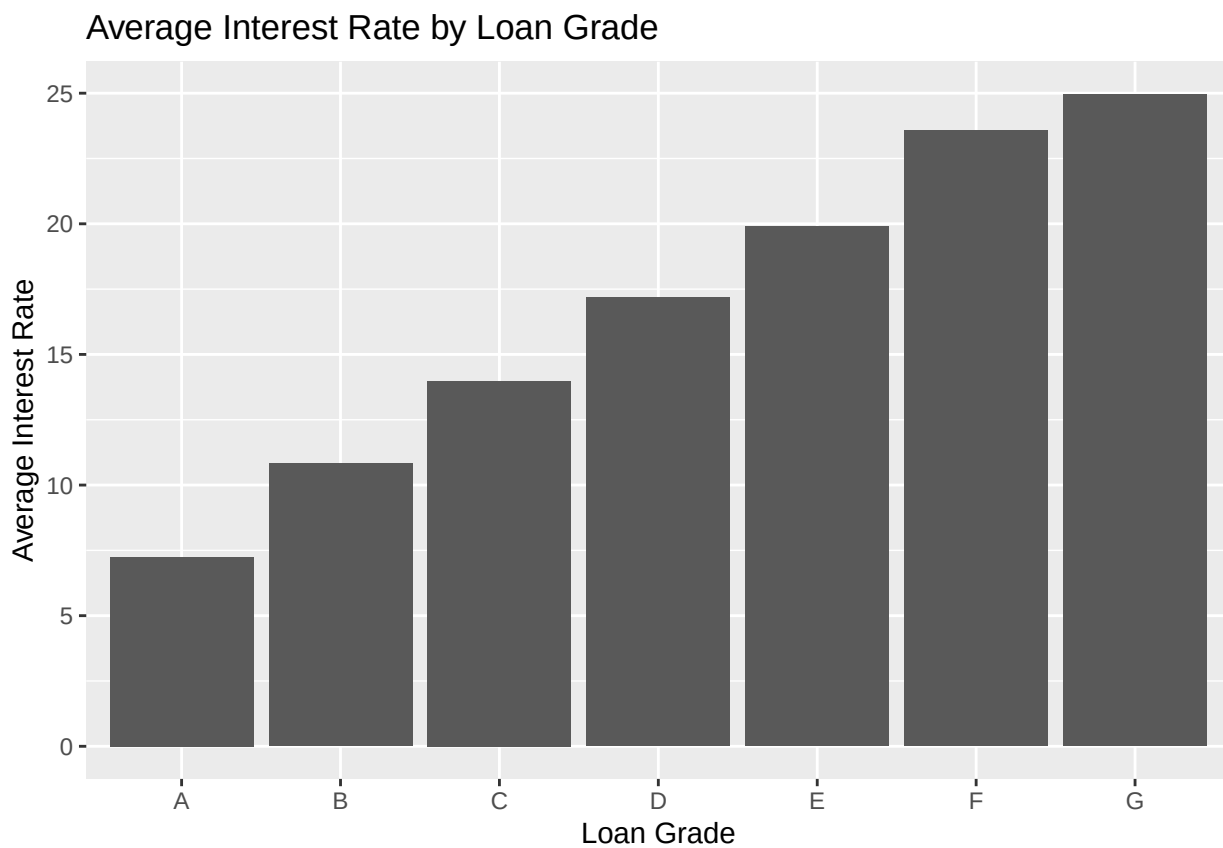
Summary:

```
summary_table <- actual_data %>%
  dplyr::group_by(grade) %>%
  dplyr::summarize(
  avg_int_rate = mean(int_rate, na.rm = TRUE)) %>%
  arrange(grade)

summary_table
```

```
## # A tibble: 7 x 2
##   grade avg_int_rate
##   <ord>        <dbl>
## 1 A             7.24
## 2 B            10.8
## 3 C            14.0
## 4 D            17.2
## 5 E            19.9
## 6 F            23.6
## 7 G            25.0
```

Graph:

```
ggplot(
  dplyr::summarize(
    dplyr::group_by(actual_data, grade),
    avg_int_rate = mean(int_rate, na.rm = TRUE)
  ),
  aes(x = grade, y = avg_int_rate)
) +
  geom_col() +
  labs(
    title = "Average Interest Rate by Loan Grade",
    x = "Loan Grade",
    y = "Average Interest Rate"
  )
```



Interpretation of Findings:

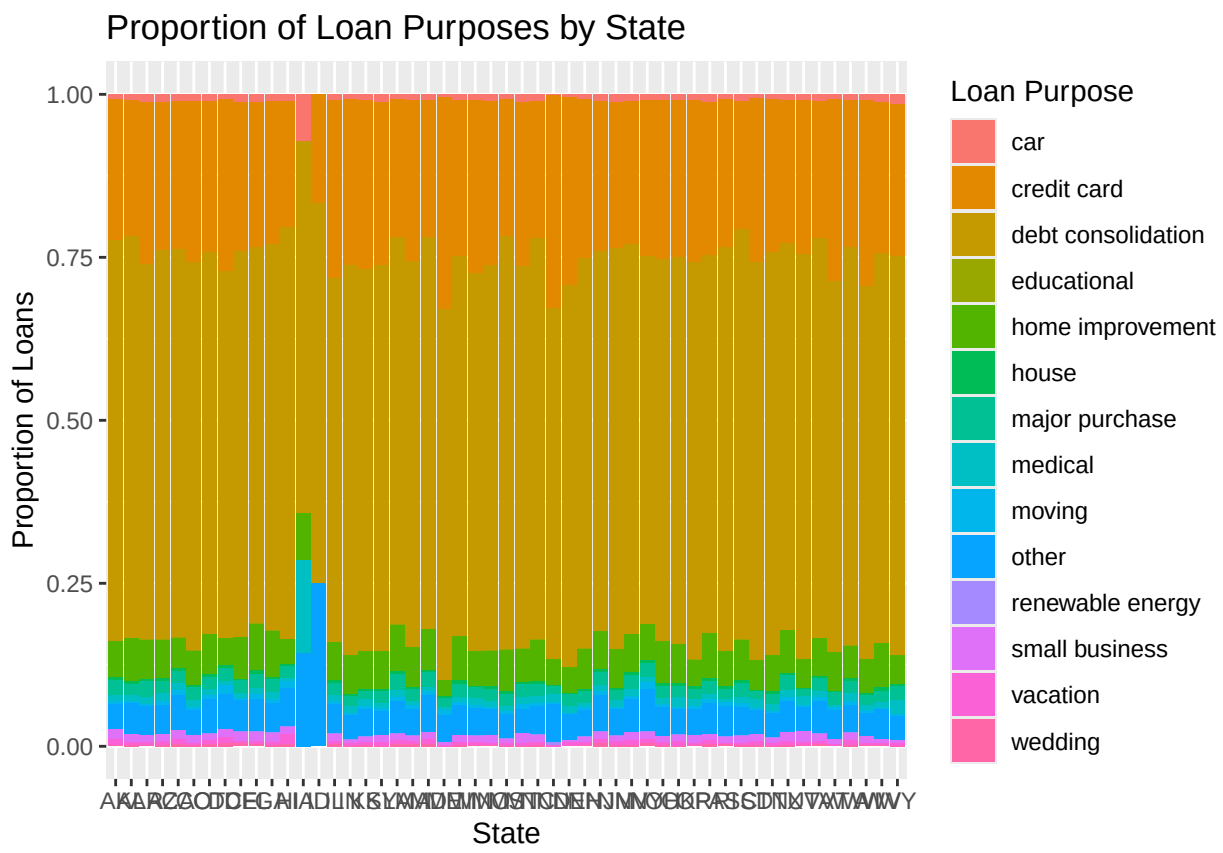
Q2: Is there a relationship between state and purpose of loan?

Summary:

```
#
```

Graph:

```
ggplot(actual_data, aes(x = addr_state, fill = purpose)) +
  geom_bar(position = "fill") +
  labs(
    title = "Proportion of Loan Purposes by State",
    x = "State",
    y = "Proportion of Loans",
    fill = "Loan Purpose"
  )
```



Interpretation of Findings:

Deliverer: Emily Jones

Q1: Does the purpose of the loan correlate to the amount of the loan?

Summary:

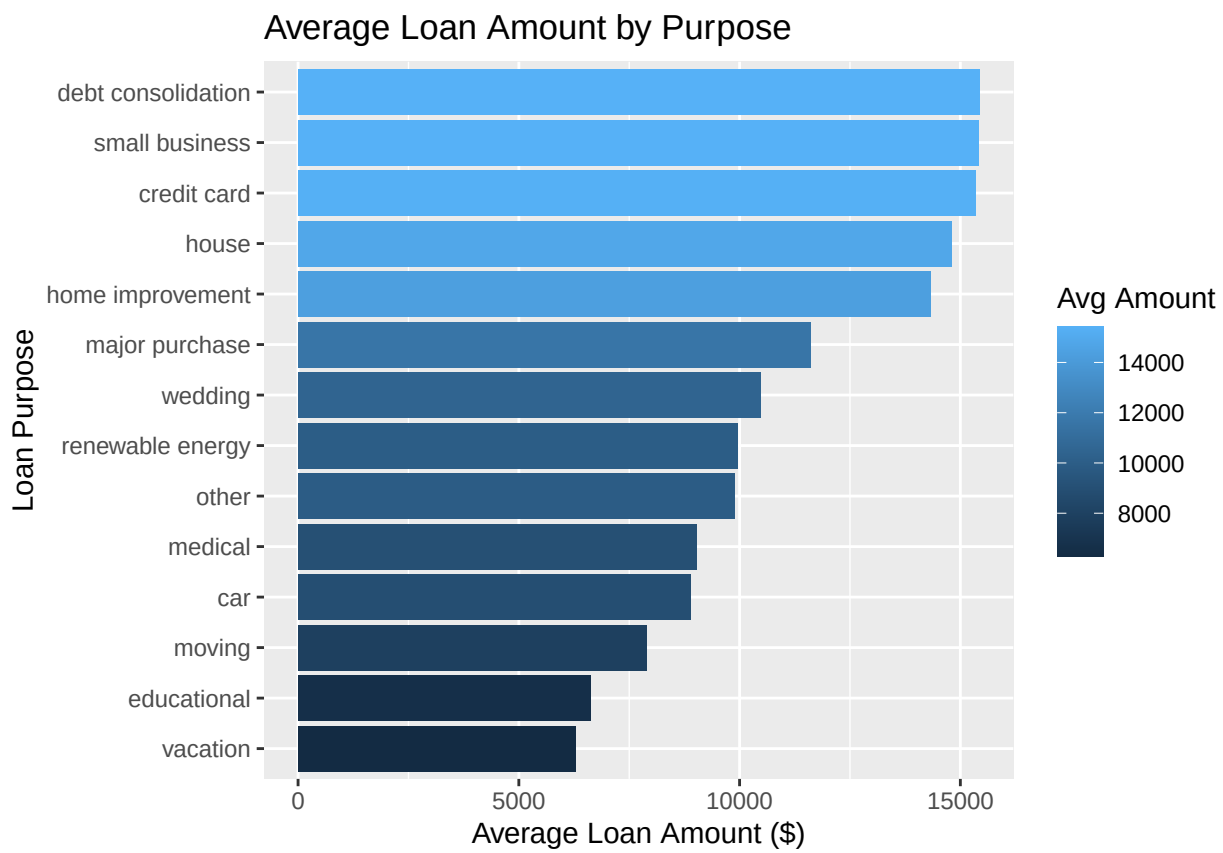
```
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

## The following object is masked from 'package:readr':
##
##     col_factor
```

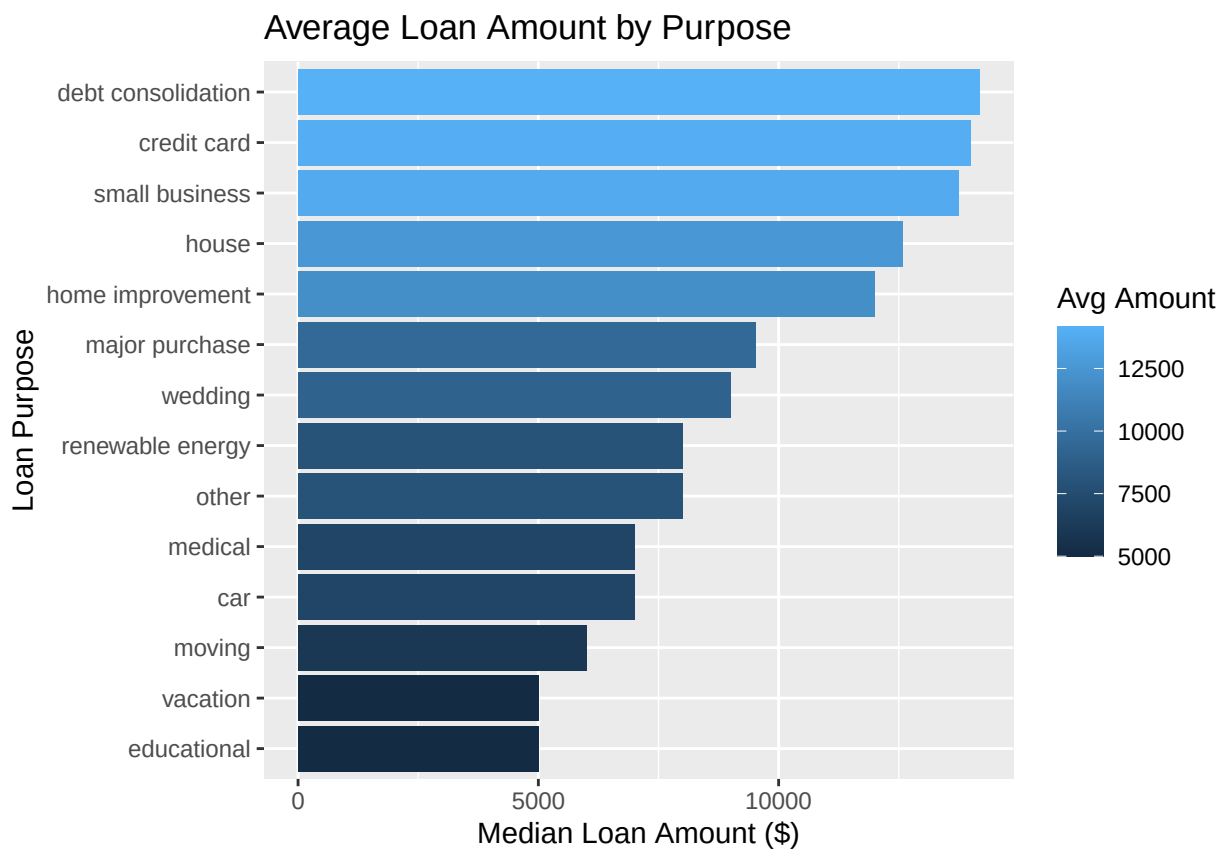
```r
loan_summary_table <- actual_data %>%
  filter(!is.na(purpose), !is.na(loan_amnt)) %>%
  group_by(purpose) %>%
  summarise(
    n = n(),
    mean_loan = mean(loan_amnt),
    median_loan = median(loan_amnt)
  ) %>%
  arrange(desc(mean_loan)) %>%
  mutate(median_loan = dollar(median_loan)) |>
  mutate(mean_loan
         = dollar(mean_loan)) %>%
  rename(
    "Loan Purpose" = purpose,
    "Number of Loans" = n,
    "Average Loan Amount" = mean_loan,
    "Median Loan Amount" = median_loan,
  ) |> gt()

loan_summary_table
```

Graph:

```r
#graph 1
bar_data <- actual_data |>
  filter(!is.na(purpose), !is.na(loan_amnt)) |>
  group_by(purpose) |>
  summarise(mean_loan = mean(loan_amnt)) |>
  arrange(mean_loan)

ggplot(bar_data, aes(x = reorder(purpose, mean_loan), y = mean_loan, fill = mean_loan)) +
  coord_flip() +
  geom_col()+
  labs(
    title = "Average Loan Amount by Purpose",
    x = "Loan Purpose",
    y = "Average Loan Amount ($)",
    fill = "Avg Amount")
```




```r
#graph 2
bar_data <- actual_data |>
  filter(!is.na(purpose), !is.na(loan_amnt)) |>
  group_by(purpose) |>
  summarise(median_loan = median(loan_amnt)) |>
  arrange(median_loan)

ggplot(bar_data, aes(x = reorder(purpose, median_loan), y = median_loan, fill = median_loan)) +
  coord_flip() +
  geom_col()+
  labs(
    title = "Average Loan Amount by Purpose",
    x = "Loan Purpose",
    y = "Median Loan Amount ($)",
    fill = "Avg Amount")
```



Interpretation of Findings: loan purpose is meaningfully associated with loan size. Debt consolidation, small business, and credit card loans sit at the top with average amounts around $15,000–$15,400, reflecting that these borrowers are addressing large existing obligations or financing business operations. Vacation and educational loans are at the bottom, averaging around $6,300–$6,600, consistent with smaller, more discretionary spending. The median rankings largely mirror the mean, with one notable exception: house loans have a higher median ($12,575) relative to their mean ($14,798), suggesting a left-skewed distribution where a cluster of smaller house loans pulls the median down.

Q2:

Summary:

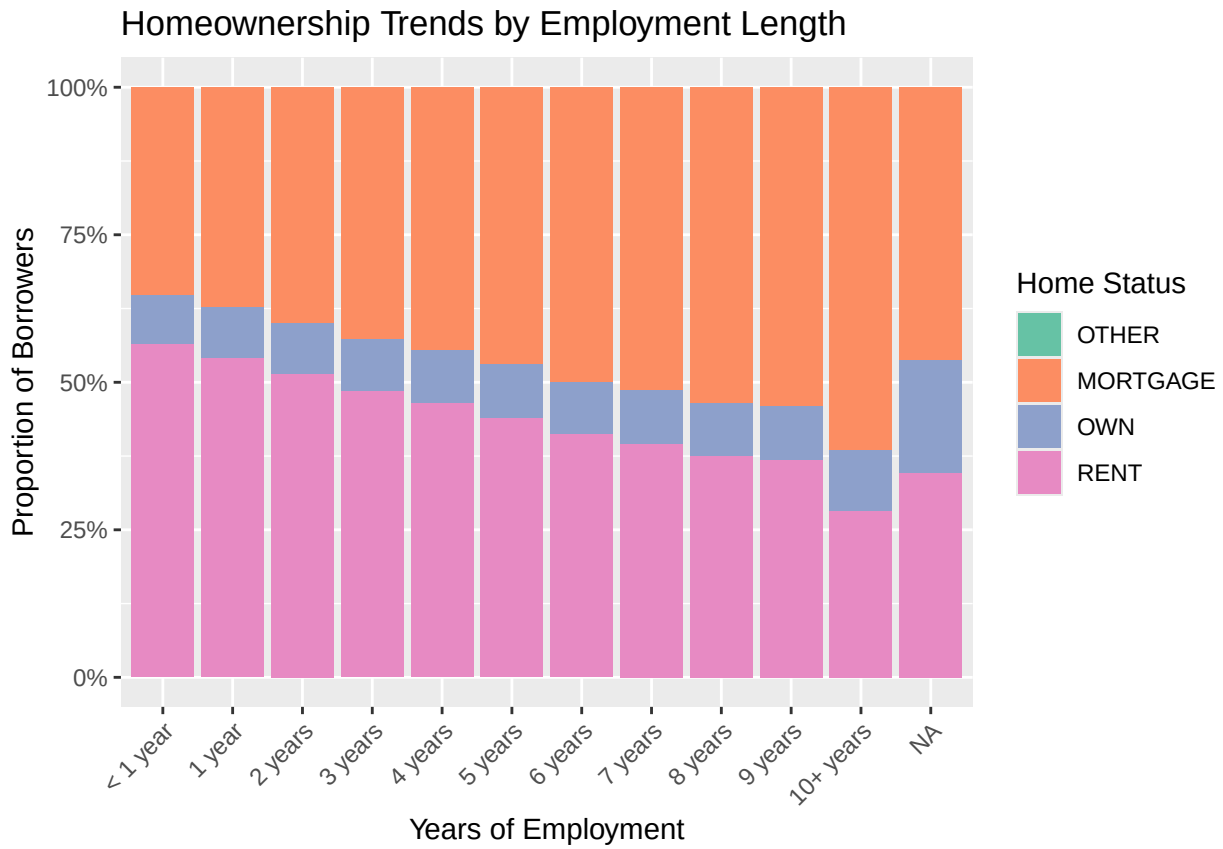
```
summary_table <- actual_data %>%
  filter(!is.na(emp_length), !is.na(home_ownership)) %>%
  group_by(emp_length, home_ownership) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(emp_length) %>%
  mutate(percentage = count / sum(count) * 100) %>%
  select(-count) %>%
  pivot_wider(names_from = home_ownership, values_from = percentage)

summary_table
```

```
## # A tibble: 11 x 5
## # Groups:   emp_length [11]
```

```
##    emp_length  OTHER MORTGAGE   OWN  RENT
##    <ord>       <dbl>    <dbl> <dbl> <dbl>
##  1 < 1 year   0.0567     35.2  8.40  56.4
##  2 1 year     0.0420     37.3  8.68  54.0
##  3 2 years    0.0330     39.9  8.64  51.5
##  4 3 years    0.0271     42.6  8.96  48.4
##  5 4 years    0.0247     44.5  9.06  46.4
##  6 5 years    0.0251     46.9  9.26  43.9
##  7 6 years    0.0373     50.0  8.86  41.2
##  8 7 years    0.0157     51.4  9.00  39.5
##  9 8 years    0.0114     53.5  9.00  37.5
## 10 9 years    0.0231     54.0  9.12  36.8
## 11 10+ years  0.0202     61.5 10.3   28.2
```

Graph:

```r
emp_levels <- c("< 1 year", "1 year", "2 years", "3 years", "4 years",
                "5 years", "6 years", "7 years", "8 years", "9 years", "10+ years")

actual_data$emp_length <- factor(actual_data$emp_length, levels = emp_levels)

ggplot(actual_data, aes(x = emp_length, fill = home_ownership)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Homeownership Trends by Employment Length",
    x = "Years of Employment",
    y = "Proportion of Borrowers",
    fill = "Home Status"
  )+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Interpretation of Findings: TThe data reveals a compelling and practically significant trend: as employment length increases, the proportion of mortgage holders rises from 35% (under 1 year) to over 61% (10+ years), while renters decline from 56% to just 28%. The proportion of outright owners holds steady at roughly 9% across all employment levels. This suggests that employment tenure is a strong proxy for financial stability and housing progression — longer-tenured workers have had more time to accumulate the savings and credit history needed to obtain a mortgage. From a credit risk standpoint, this is an actionable insight: short-tenure renters, who make up a disproportionate share of early-career borrowers, may represent higher default risk, not because of their purpose or income level, but because of their overall financial instability profile. LendingClub should consider employment length as a meaningful input into its risk scoring model.

Follow-up Questions

New Questions Based Off Initial Investigation

- Q1: Which borrower and loan characteristics are the most significant predictors of interest rate, and how does model performance change as we add complexity?
- Q2: Across the five states with the highest loan volume (CA, NY, TX, FL, IL), do the significant predictors of interest rate remain consistent, or do certain borrower characteristics become more or less predictive depending on the state?
- Q3: Within Grade B and C loans (which make up over 56% of the portfolio), what combination of DTI, employment length, and home ownership best separates defaulters from non-defaulters?
- Q4: Are 60-month term loans more likely to default than 36-month loans within the same grade, and does this gap widen at lower grades (D–G)?

Investigation of Follow-up Questions

GIVE WHAT 2 QUESTIONS YOU ATTEMPTED TO INVESTIGATE FURTHER IN COMPLETE SENTENCES (*Example:* Our group decided to investigate Q2 and Q4 in further detail.)

SHOW AT LEAST 2 TABLES OR FIGURES BELOW THAT EXPLORE ANSWERS FOR THE QUESTIONS YOU ARE INVESTIGATING FURTHER.

#Q1: Which borrower and loan characteristics are the most significant predictors of interest rate, and how does model performance change as we add complexity?

```
q1_data <- actual_data |>
  drop_na(int_rate, dti, annual_inc, loan_amnt, term, home_ownership, purpose)

m0 <- lm(int_rate ~ 1,                                                          data = q1_data)
m1 <- lm(int_rate ~ dti,                                                        data = q1_data)
m2 <- lm(int_rate ~ annual_inc,                                                 data = q1_data)
m3 <- lm(int_rate ~ loan_amnt,                                                  data = q1_data)
m4 <- lm(int_rate ~ dti + annual_inc,                                           data = q1_data)
m5 <- lm(int_rate ~ dti + annual_inc + loan_amnt,                               data = q1_data)
m6 <- lm(int_rate ~ dti + annual_inc + loan_amnt + term,                        data = q1_data)
m7 <- lm(int_rate ~ dti + annual_inc + loan_amnt + term + home_ownership,       data = q1_data)
m8 <- lm(int_rate ~ dti + annual_inc + loan_amnt + term + home_ownership + purpose, data = q1_data)
```

Table 1: Sequential ANOVA results

```
anova(m0, m1, m4, m5, m6, m7, m8) |>
  tidy() |>
  mutate(
    Predictors = c(
      "Intercept only",
      "+ dti",
      "+ annual_inc",
      "+ loan_amnt",
      "+ term (1 dummy)",
      "+ home_ownership (5 dummies)",
      "+ purpose (13 dummies)"
    ),
    rss      = round(rss, 0),
    sumsq    = round(sumsq, 0),
    statistic = round(statistic, 1),
    p.value  = ifelse(p.value < 0.001, "< 2.2e-16 ***", round(p.value, 4))
  ) |>
  select(Predictors, df.residual, rss, df, sumsq, statistic, p.value) |>
  rename(
    `Res. Df`     = df.residual,
    RSS           = rss,
    Df            = df,
    `Sum of Sq`   = sumsq,
    `F Statistic` = statistic,
    `P-Value`     = p.value
  ) |>
  gt() |>
  tab_header(
```

```r
    title    = "Sequential Nested F-Tests: Adding Predictors to Interest Rate Model",
    subtitle = "Each row tests whether adding that predictor significantly reduces unexplained error"
  )
```

Table 2: Type II ANOVA. Tests each predictor controlling for all others

```r
Anova(m8, type = 2) |>
  tidy() |>
  mutate(
    sumsq     = round(sumsq, 0),
    statistic = round(statistic, 1),
    p.value   = ifelse(p.value < 0.001, "< 2.2e-16 ***", round(p.value, 4))
  ) |>
  filter(term != "Residuals") |>
  rename(
    Predictor    = term,
    `Sum Sq`     = sumsq,
    Df           = df,
    `F Statistic` = statistic,
    `P-Value`    = p.value
  ) |>
  arrange(desc(`F Statistic`)) |>
  gt() |>
  tab_header(
    title    = "Type II ANOVA: Full Model (M8)",
    subtitle = "Each predictor tested while controlling for all others simultaneously"
  )
```

Figure 1: RSS Drop at Each Step

```r
rss_data <- tibble(
  Model = c("M0\n(intercept)", "M1\n+dti", "M4\n+annual_inc",
            "M5\n+loan_amnt", "M6\n+term", "M7\n+home_own", "M8\n+purpose"),
  RSS   = c(
    sum(residuals(m0)^2), sum(residuals(m1)^2),
    sum(residuals(m4)^2), sum(residuals(m5)^2),
    sum(residuals(m6)^2), sum(residuals(m7)^2),
    sum(residuals(m8)^2)
  ),
  Step = 1:7
)

ggplot(rss_data, aes(x = reorder(Model, Step), y = RSS / 1e6, group = 1)) +
  geom_line(color = "red", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  geom_text(aes(label = paste0(round(RSS / 1e6, 1), "M")),
            vjust = -0.8, size = 3.5) +
  annotate("rect",
           xmin = 4.5, xmax = 5.5,
           ymin = -Inf, ymax = Inf,
           alpha = 0.1, fill = "blue") +
  annotate("text", x = 5, y = 16,
           label = "Largest\ndrop", size = 3.5,
```

```
            color = "blue", fontface = "bold") +
  labs(
    title    = "Loan term dominates interest rate prediction, adding it halves unexplained error",
    subtitle = "Each step tests whether the new predictor significantly reduces RSS via nested",
    x        = "Model",
    y        = "RSS (millions)"
  ) +
  theme_minimal()
```

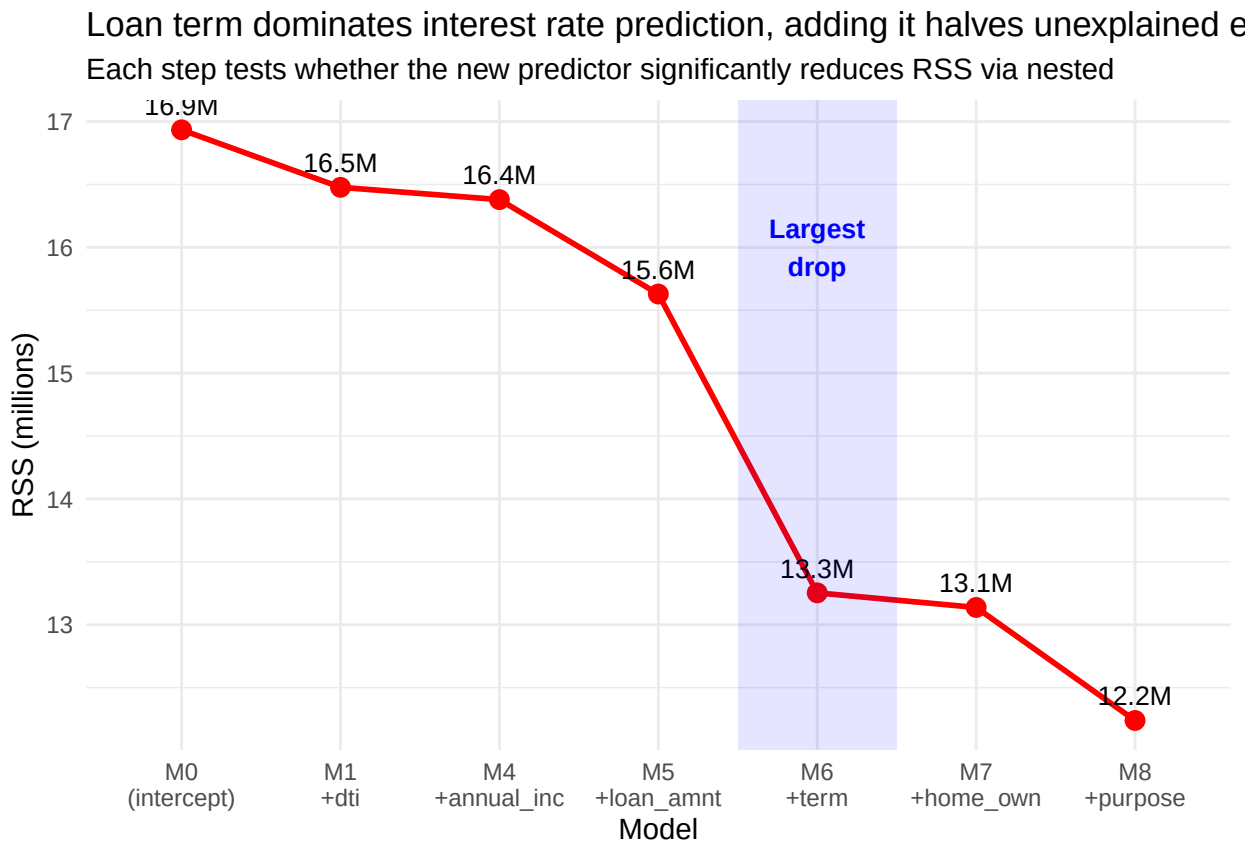
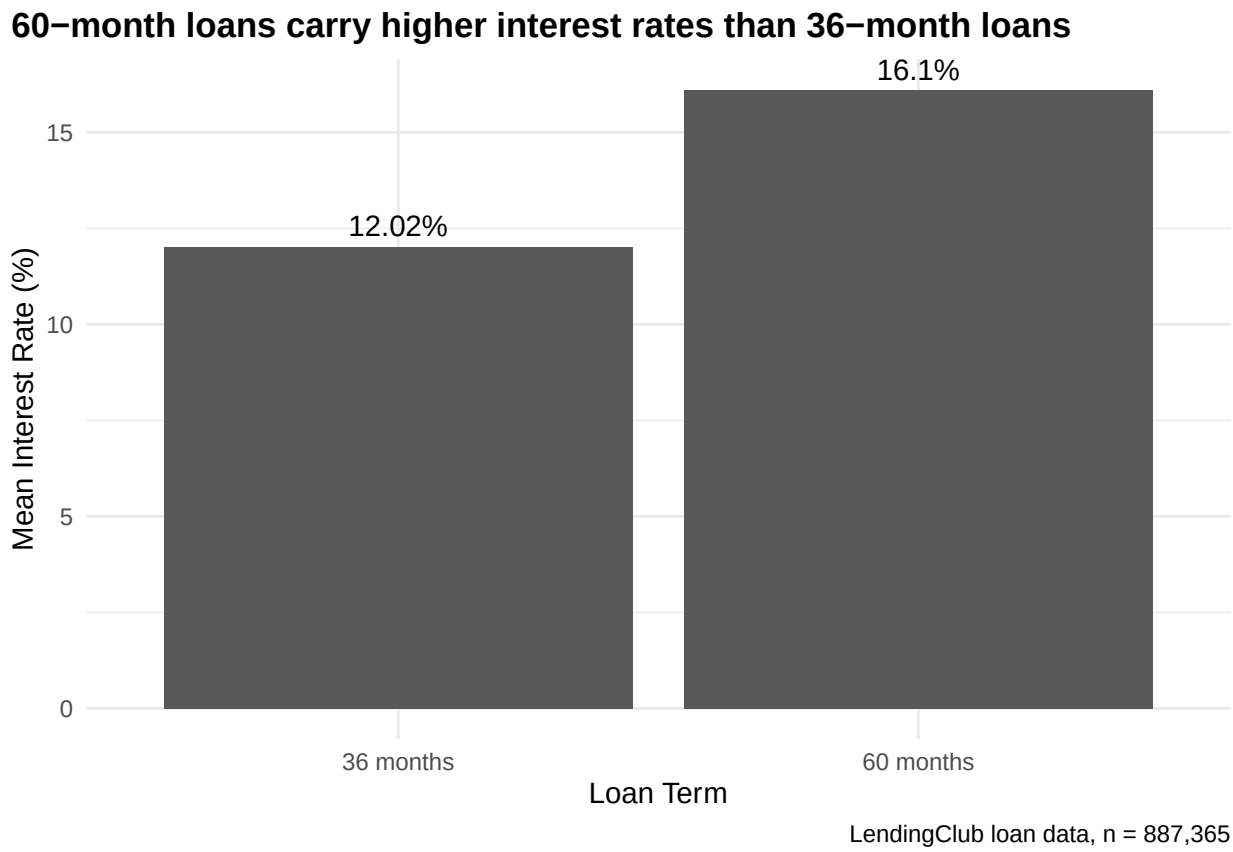


Figure 2: Mean Interest Rate by Term (most significant predictor)

```
actual_data |>
  group_by(term) |>
  summarise(
    mean_int_rate = round(mean(int_rate, na.rm = TRUE), 2),
    .groups = "drop"
  ) |>
  ggplot(aes(x = term, y = mean_int_rate)) +
  geom_col() +
  geom_text(aes(label = paste0(mean_int_rate, "%")),
            vjust = -0.5) +
  labs(
    title   = "60-month loans carry higher interest rates than 36-month loans",
    x       = "Loan Term",
    y       = "Mean Interest Rate (%)",
    caption = "LendingClub loan data, n = 887,365"
```

```
  ) +
  theme_minimal() +
  theme(
    plot.title          = element_text(face = "bold"),
    plot.title.position = "plot"
  )
```



#Q2: Across the five states with the highest loan volume (CA, NY, TX, FL, IL), do the significant predictors of loan grade remain consistent, or do certain borrower characteristics become more or less predictive depending on the state?

```
#
```

Summary

GIVE A 2 PARAGRAPH SUMMARY.

PARAGRAPH 1 SHOULD DESCRIBE WHAT YOU LEARNED ABOUT YOUR DATA FROM INVESTIGATING THE INITIAL QUESTIONS. DID YOU FIND ANYTHING UNUSUAL IN YOUR DATA? DID ANYTHING SURPRISE YOU? WHICH OF THE INITIAL QUESTIONS WERE HELPFUL IN LEADING YOU TO MORE QUESTIONS?

PARAGRAPH 2 SHOULD SUMMARIZE WHAT YOU LEARNED FROM INVESTIGATING THE FOLLOW-UP QUESTIONS. WHY ARE THESE FOLLOW-UP QUESTIONS INTERESTING FOR INVESTIGATION? DESCRIBE THE TABLES/FIGURES YOU USED TO EXPLORE ANSWERS

TO THESE FOLLOW-UP QUESTIONS? WHAT DID YOU LEARN FROM THE TABLES/FIGURES REGARDING THE FOLLOW-UP QUESTIONS YOU PROPOSED?

home_ownership	inc_decile	n	inc_min	inc_max	mean_inc
OTHER	1	33	10000	34000	26265
OTHER	2	37	35000	42000	38284
OTHER	3	27	42162	50000	46088
OTHER	4	13	50000	55000	52868
OTHER	5	21	56004	64000	59800
OTHER	6	21	65000	72000	68691
OTHER	7	27	72204	83000	78143
OTHER	8	28	84000	98000	90137
OTHER	9	13	100000	120000	106731
OTHER	10	11	125000	157500	149319
OTHER	NA	4	Inf	-Inf	NaN
MORTGAGE	1	19481	3800	34000	28256
MORTGAGE	2	29449	34000	42000	38425
MORTGAGE	3	36121	42000	50000	45961
MORTGAGE	4	40529	50000	56000	52614
MORTGAGE	5	45205	56000	65000	60121
MORTGAGE	6	48169	65000	72000	68213
MORTGAGE	7	51417	72000	83500	77743
MORTGAGE	8	54950	83500	98500	90263
MORTGAGE	9	57615	98500	124800	109248
MORTGAGE	10	60612	124800	157500	146878
OWN	1	14420	4000	34000	25887
OWN	2	10869	34000	42000	38103
OWN	3	9223	42000	50000	45774
OWN	4	8857	50000	56000	52409
OWN	5	8045	56000	65000	59996
OWN	6	7541	65000	72000	68050
OWN	7	7223	72000	83500	77569
OWN	8	6856	83500	98500	90074
OWN	9	6925	98500	124800	108982
OWN	10	7511	124800	157500	147326
RENT	1	54803	1896	34000	26490
RENT	2	48381	34000	42000	38190
RENT	3	43365	42000	50000	45807
RENT	4	39337	50000	56000	52463
RENT	5	35465	56000	65000	60018
RENT	6	33005	65000	72000	68070
RENT	7	30069	72000	83500	77609
RENT	8	26902	83500	98500	89966
RENT	9	24183	98500	124800	108822
RENT	10	20602	124800	157500	146108

grade	n	mean_loan_amnt	median_loan_amnt	sd_loan_amnt	q1	q3
A	148202	14038.86	12000	7502.13	8000	20000
B	254532	13642.96	12000	7761.08	8000	18000
C	245858	14466.39	12000	8538.68	8000	20000
D	139536	15459.54	14000	9012.78	8500	21000
E	70702	17998.06	16850	9196.25	11000	25000
F	23046	19139.16	18625	9213.62	12000	25875
G	5489	20538.56	20000	8789.57	13475	27550

purpose	n	mean_dti	median_dti	sd_dti	q1	q3
car	8863	14.36	13.45	8.56	7.70	20.16
credit card	206180	18.33	17.78	8.07	12.21	23.97
debt consolidation	524205	18.77	18.32	8.16	12.67	24.50
educational	423	11.26	11.01	6.84	5.44	16.28
home improvement	51828	15.55	14.71	8.54	9.03	21.32
house	3707	14.36	13.61	8.52	7.64	20.20
major purchase	17277	14.60	13.62	8.66	7.85	20.47
medical	8540	17.23	16.73	8.80	10.62	23.36
moving	5414	16.14	15.55	8.64	9.61	22.13
other	42893	16.84	16.33	8.74	10.18	22.93
renewable energy	575	15.72	15.21	8.37	9.44	21.52
small business	10377	14.03	13.13	8.30	7.54	19.64
vacation	4736	17.14	16.72	8.62	10.44	23.03
wedding	2347	13.99	13.75	7.57	8.21	19.60

Loan Purpose	Number of Loans	Average Loan Amount	Median Loan Amount
debt consolidation	524205	$15,428.48	$14,175
small business	10377	$15,420.43	$13,750
credit card	206180	$15,338.93	$14,000
house	3707	$14,798.10	$12,575
home improvement	51828	$14,321.11	$12,000
major purchase	17277	$11,600.00	$9,525
wedding	2347	$10,468.63	$9,000
renewable energy	575	$9,948.30	$8,000
other	42893	$9,889.14	$8,000
medical	8540	$9,019.07	$7,000
car	8863	$8,882.43	$7,000
moving	5414	$7,891.57	$6,000
educational	423	$6,616.08	$5,000
vacation	4736	$6,289.54	$5,000

Sequential Nested F-Tests: Adding Predictors to Interest Rate Model

Each row tests whether adding that predictor significantly reduces unexplained error

Predictors	Res. Df	RSS	Df	Sum of Sq	F Statistic	P-Value
Intercept only	887360	16934478	NA	NA	NA	NA
+ dti	887359	16477466	1	457012	33133.9	< 2.2e-16 ***
+ annual_inc	887358	16379754	1	97712	7084.2	< 2.2e-16 ***
+ loan_amnt	887357	15629242	1	750512	54413.0	< 2.2e-16 ***
+ term (1 dummy)	887356	13253483	1	2375759	172245.3	< 2.2e-16 ***
+ home_ownership (5 dummies)	887353	13137505	3	115977	2802.8	< 2.2e-16 ***
+ purpose (13 dummies)	887340	12238976	13	898529	5011.1	< 2.2e-16 ***

Type II ANOVA: Full Model (M8)

Each predictor tested while controlling for all others simultaneously

Predictor	Sum Sq	Df	F Statistic	P-Value
term	2283112	1	165528.3	< 2.2e-16 ***
annual_inc	286171	1	20747.7	< 2.2e-16 ***
dti	129433	1	9384.1	< 2.2e-16 ***
loan_amnt	91450	1	6630.3	< 2.2e-16 ***
purpose	898529	13	5011.1	< 2.2e-16 ***
home_ownership	101973	3	2464.4	< 2.2e-16 ***